Note for Integrating Factors:

We can take the integrating factor to be any non-zero constant multiple of an integrating factor. For example, $e^{\int \frac{1}{x} dx} = e^{\ln|x|} = |x|$

On any interval not containing $x = 0$, $|x|$ differs from x only by a constant factor (namely ± 1), so we may take the integrating factor to be x. So throughout these worked solutions, the modulus signs will be dropped when multiplying the differential equation through by the integrating factor.

1) a)

$\frac{d}{dx}(6xy) = 6x \times \frac{dy}{dx} + 6 \times y$

$= 6x\frac{dy}{dx} + 6y$

1) b)

$\frac{d}{dx}(e^{3x}y) = e^{3x} \times \frac{dy}{dx} + 3e^{3x} \times y$

$= e^{3x}\frac{dy}{dx} + 3e^{3x}y$

1) c)

$\frac{d}{dx}(y \tan 4x) = y \times 4\sec^2 4x + \frac{dy}{dx} \times \tan 4x$

$= \tan 4x\frac{dy}{dx} + 4y\sec^2 4x$

1) d)

$\frac{d}{dx}(y \ln 2x) = y \times \frac{1}{x} + \frac{dy}{dx} \times \ln 2x$

$= \ln 2x\frac{dy}{dx} + \frac{y}{x}$

2) a)

$\frac{d}{dx}(yx^2) = y \times 2x + \frac{dy}{dx} \times x^2$

$= x^2\frac{dy}{dx} + 2xy$

2) b) i)

$x^2\frac{\mathrm{d}y}{\mathrm{d}x}+2xy=4$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(yx^2)=4$

$\Rightarrow yx^2=4x+c$

$\Rightarrow y=\frac{4x+c}{x^2}$

2) b) ii)

$x^2\frac{\mathrm{d}y}{\mathrm{d}x}+2xy=4x$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(yx^2)=4x$

$\Rightarrow yx^2=2x^2+c$

$\Rightarrow y=\frac{2x^2+c}{x^2}$

3) a)

$\frac{\mathrm{d}}{\mathrm{d}x}(y\sin x)=y\times\cos x+\frac{\mathrm{d}y}{\mathrm{d}x}\times\sin x$

$=\sin x\frac{\mathrm{d}y}{\mathrm{d}x}+y\cos x$

3) b) i)

$\sin x\frac{\mathrm{d}y}{\mathrm{d}x}+y\cos x=3x^2$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(y\sin x)=3x^2$

$\Rightarrow y\sin x=x^3+c$

$\Rightarrow y=\frac{x^3+c}{\sin x}$

3) b) ii)

$\sin x\frac{\mathrm{d}y}{\mathrm{d}x}+y\cos x=\mathrm{e}^x$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(y\sin x)=\mathrm{e}^x$

$\Rightarrow y\sin x=\mathrm{e}^x+c$

$\Rightarrow y=\frac{\mathrm{e}^x+c}{\sin x}$

4) a)

$\frac{\mathrm{d}}{\mathrm{d}x}(\mathrm{e}^{5x}y) = \mathrm{e}^{5x} \times \frac{\mathrm{d}y}{\mathrm{d}x} + 5\mathrm{e}^{5x} \times y$

$= \mathrm{e}^{5x}\frac{\mathrm{d}y}{\mathrm{d}x} + 5\mathrm{e}^{5x}y$

4) b)

$\mathrm{e}^{5x}\frac{\mathrm{d}y}{\mathrm{d}x} + 5\mathrm{e}^{5x}y = 6\mathrm{e}^{3x}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(\mathrm{e}^{5x}y) = 6\mathrm{e}^{3x}$

$\Rightarrow \mathrm{e}^{5x}y = 2\mathrm{e}^{3x} + c$

$\Rightarrow y = \frac{2\mathrm{e}^{3x}+c}{\mathrm{e}^{5x}}$

4) c)

$0 = \frac{2\mathrm{e}^{3\ln 2}+c}{\mathrm{e}^{5\ln 2}}$

$\Rightarrow 0 = \frac{2\mathrm{e}^{\ln 2^3}+c}{\mathrm{e}^{\ln 2^5}}$

$\Rightarrow 0 = \frac{2\mathrm{e}^{\ln 8}+c}{\mathrm{e}^{\ln 32}}$

$\Rightarrow 0 = \frac{2\times 8+c}{32}$

$\Rightarrow 0 = 16 + c$

$\Rightarrow c = -16$

$\therefore y = \frac{2\mathrm{e}^{3x}-16}{\mathrm{e}^{5x}}$

5) a)

$x^4\frac{\mathrm{d}y}{\mathrm{d}x} + 4x^3y = x^8$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(x^4y) = x^8$

$\Rightarrow x^4y = \frac{1}{9}x^9 + c$

$\Rightarrow y = \frac{1}{9}x^5 + cx^{-4}$

5) b)

$\cos 2x \frac{\mathrm{d}y}{\mathrm{d}x} - 2y \sin 2x = \cos 2x$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(y \cos 2x) = \cos 2x$

$\Rightarrow y \cos 2x = \frac{1}{2}\sin 2x + c$

$\Rightarrow y = \frac{\sin 2x}{2 \cos 2x} + \frac{c}{\cos 2x}$

$\Rightarrow y = \frac{1}{2}\tan 2x + c \sec 2x$

5) c)

$\tan 3x \frac{\mathrm{d}y}{\mathrm{d}x} + 3y \sec^2 3x = \cos 3x$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(y \tan 3x) = \cos 3x$

$\Rightarrow y \tan 3x = \frac{1}{3}\sin 3x + c$

$\Rightarrow y = \frac{\sin 3x}{3 \tan 3x} + \frac{c}{\tan 3x}$

$\Rightarrow y = \frac{1}{3}\sin 3x \times \frac{\cos 3x}{\sin 3x} + c \cot 3x$

$\Rightarrow y = \frac{1}{3}\cos 3x + c \cot 3x$

5) d)

$\mathrm{e}^{-9x} \frac{\mathrm{d}y}{\mathrm{d}x} - 9\mathrm{e}^{-9x} y = \mathrm{e}^{-8x}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(\mathrm{e}^{-9x} y) = \mathrm{e}^{-8x}$

$\Rightarrow \mathrm{e}^{-9x} y = -\frac{1}{8}\mathrm{e}^{-8x} + c$

$\Rightarrow y = -\frac{1}{8}\mathrm{e}^{x} + c\mathrm{e}^{9x}$

6) a)

$\mathrm{e}^{\int 2x \mathrm{d}x}$

$= \mathrm{e}^{x^2}$

6) b)

$\mathrm{e}^{\int \frac{1}{x} \mathrm{d}x}$

$= \mathrm{e}^{\ln|x|}$

$= |x|$

6) c)

$e^{\int \frac{1}{2x+1} dx}$

$= e^{\frac{1}{2}\ln|2x+1|}$

$= e^{\ln\left|(2x+1)^{\frac{1}{2}}\right|}$

$= \left|(2x+1)^{\frac{1}{2}}\right|$

$= (2x+1)^{\frac{1}{2}}$

6) d)

$e^{\int \tan x dx}$

$e^{\int \frac{\sin x}{\cos x} dx}$

$= e^{-\ln|\cos x|}$

$= e^{\ln|(\cos x)^{-1}|}$

$= e^{\ln|\sec x|}$

$= |\sec x|$

7) a)

Integrating Factor

$= e^{\int 8 dx}$

$= e^{8x}$

7) b)

Integrating Factor

$= e^{\int (-3) dx}$

$= e^{-3x}$

7) c)

Integrating Factor

$= e^{\int (-x) dx}$

$= e^{-\frac{1}{2}x^2}$

7) d)

Integrating Factor

$= \mathrm{e}^{\int (2x+1)\mathrm{d}x}$

$= \mathrm{e}^{x^2+x}$

8) a)

$x^3 \frac{\mathrm{d}y}{\mathrm{d}x} + x^3 y = \frac{2}{\mathrm{e}^x}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} + y = \frac{2}{x^3 \mathrm{e}^x}$

Integrating Factor

$= \mathrm{e}^{\int 1\mathrm{d}y}$

$= \mathrm{e}^x$

8) b)

$\frac{\mathrm{d}y}{\mathrm{d}x} \times \mathrm{e}^x + y \times \mathrm{e}^x = \frac{2}{x^3 \mathrm{e}^x} \times \mathrm{e}^x$

$\Rightarrow \mathrm{e}^x \frac{\mathrm{d}y}{\mathrm{d}x} + y\mathrm{e}^x = \frac{2}{x^3}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(y\mathrm{e}^x) = 2x^{-3}$

$\Rightarrow y\mathrm{e}^x = -x^{-2} + C$

$\Rightarrow y = -\frac{1}{x^2 \mathrm{e}^x} + \frac{C}{\mathrm{e}^x}$

8) c) i)

$\frac{\mathrm{d}y}{\mathrm{d}x} \times 2\mathrm{e}^x + y \times 2\mathrm{e}^x = \frac{2}{x^3 \mathrm{e}^x} \times 2\mathrm{e}^x$

$\Rightarrow 2\mathrm{e}^x \frac{\mathrm{d}y}{\mathrm{d}x} + 2y\mathrm{e}^x = \frac{4}{x^3}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(2y\mathrm{e}^x) = 4x^{-3}$

$\Rightarrow 2y\mathrm{e}^x = -2x^{-2} + c_1$

$\Rightarrow y = -\frac{2}{x^2 \times 2\mathrm{e}^x} + \frac{c_1}{2\mathrm{e}^x}$

$\Rightarrow y = -\frac{1}{x^2 \mathrm{e}^x} + \frac{c_1}{2\mathrm{e}^x}$

This is the same as the answer in b), where $\frac{c_1}{2} = C$

8) c) ii)

$\frac{\mathrm{d}y}{\mathrm{d}x} \times \mathrm{e}^{x+2} + y \times \mathrm{e}^{x+2} = \frac{2}{x^3\mathrm{e}^x} \times \mathrm{e}^{x+2}$

$\Rightarrow \mathrm{e}^{x+2}\frac{\mathrm{d}y}{\mathrm{d}x} + y\mathrm{e}^{x+2} = \frac{2\mathrm{e}^{x+2}}{x^3\mathrm{e}^x}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(y\mathrm{e}^{x+2}) = \frac{2\mathrm{e}^x \times \mathrm{e}^2}{x^3\mathrm{e}^x}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(y\mathrm{e}^{x+2}) = 2\mathrm{e}^2 x^{-3}$

$\Rightarrow y\mathrm{e}^{x+2} = -\mathrm{e}^2 x^{-2} + c_2$

$\Rightarrow y\mathrm{e}^{x+2} = -\frac{\mathrm{e}^2}{x^2} + c_2$

$\Rightarrow y = -\frac{\mathrm{e}^2}{x^2\mathrm{e}^{x+2}} + \frac{c_2}{\mathrm{e}^{x+2}}$

$\Rightarrow y = -\frac{\mathrm{e}^2}{x^2\mathrm{e}^x \times \mathrm{e}^2} + \frac{c_2}{\mathrm{e}^x \times \mathrm{e}^2}$

$\Rightarrow y = -\frac{1}{x^2\mathrm{e}^x} + \frac{c_2}{\mathrm{e}^x \times \mathrm{e}^2}$

This is the same as the answer in b), where $\frac{c_2}{\mathrm{e}^2} = C$

9) a)

$\frac{\mathrm{d}}{\mathrm{d}x}(\mu(x)y) = \mu(x)\frac{\mathrm{d}y}{\mathrm{d}x} + \mu'(x)y$

For the left hand side of the equation to equal the right hand side of the equation, we must have:

$\mu(x)\frac{\mathrm{d}y}{\mathrm{d}x} + \mu(x)\mathrm{P}(x)y = \mu(x)\frac{\mathrm{d}y}{\mathrm{d}x} + \mu'(x)y$

$\Rightarrow \mu(x)\mathrm{P}(x)y = \mu'(x)y$

$\Rightarrow \mu'(x) = \mathrm{P}(x)\mu(x)$ as required

9) b)

$\mu'(x) = \mathrm{P}(x)\mu(x)$

$\Rightarrow \frac{\mu'(x)}{\mu(x)} = \mathrm{P}(x)$

Integrating both sides with respect to x:

$\Rightarrow \int \frac{\mu'(x)}{\mu(x)}\mathrm{d}x = \int \mathrm{P}(x)\,\mathrm{d}x$

$\Rightarrow \ln|\mu(x)| = \int \mathrm{P}(x)\,\mathrm{d}x$

$\Rightarrow \mu(x) = \mathrm{e}^{\int \mathrm{P}(x)\mathrm{d}x}$

10) a)

When calculating the integrating factor, Archie should instead have got:

$e^{\int -3dx} = e^{-3x}$

10) b)

The integrating factor $\mu(x)$ must satisfy $\mu'(x) = P(x)\mu(x)$

In this example $\mu(x) = -3e^{-3x}$ and $P(x) = -3$

$\therefore 9e^{-3x} = -3 \times -3e^{-3x}$

So Archie's integrating factor $-3e^{-3x}$ does satisfy the required condition, and is therefore a valid integrating factor, even though it is incorrect based on his workings.

10) c)

Archie's solution is $y = e^{3x}\left(x - \frac{c}{3}\right)$

The other student's solution is $y = e^{3x}(x + c)$

If we let $A = -\frac{c}{3}$ then as c is an arbitrary constant, so is A

Then Archie's solution becomes $y = e^{3x}(x + A)$, which is exactly the same form as the other student's solution.

11) a)

Integrating Factor

$= e^{\int -10dx}$

$= e^{-10x}$

$\frac{dy}{dx} - 10y = e^{2x}$

$\Rightarrow e^{-10x}\frac{dy}{dx} - 10e^{-10x}y = e^{2x} \times e^{-10x}$

$\Rightarrow e^{-10x}\frac{dy}{dx} - 10e^{-10x}y = e^{-8x}$

$\Rightarrow \frac{d}{dx}(e^{-10x}y) = e^{-8x}$

$\Rightarrow e^{-10x}y = -\frac{1}{8}e^{-8x} + c$

$\Rightarrow y = -\frac{1}{8}e^{2x} + ce^{10x}$

11) b)

Integrating Factor

$= \mathrm{e}^{\int 7\mathrm{d}x}$

$= \mathrm{e}^{7x}$

$\frac{\mathrm{d}y}{\mathrm{d}x} + 7y = 8$

$\Rightarrow \mathrm{e}^{7x}\frac{\mathrm{d}y}{\mathrm{d}x} + 7\mathrm{e}^{7x}y = 8\mathrm{e}^{7x}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(\mathrm{e}^{7x}y) = 8\mathrm{e}^{7x}$

$\Rightarrow \mathrm{e}^{7x}y = \frac{8}{7}\mathrm{e}^{7x} + c$

$\Rightarrow y = \frac{8}{7} + c\mathrm{e}^{-7x}$

11) c)

Integrating Factor

$= \mathrm{e}^{\int -8\mathrm{d}x}$

$= \mathrm{e}^{-8x}$

$\frac{\mathrm{d}y}{\mathrm{d}x} - 8y = 100$

$\Rightarrow \mathrm{e}^{-8x}\frac{\mathrm{d}y}{\mathrm{d}x} - 8\mathrm{e}^{-8x}y = 100\mathrm{e}^{-8x}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(\mathrm{e}^{-8x}y) = 100\mathrm{e}^{-8x}$

$\Rightarrow \mathrm{e}^{-8x}y = -\frac{25}{2}\mathrm{e}^{-8x} + c$

$\Rightarrow y = -\frac{25}{2} + c\mathrm{e}^{8x}$

11) d)

Integrating Factor

$= \mathrm{e}^{\int 20 \mathrm{d}x}$

$= \mathrm{e}^{20x}$

$\frac{\mathrm{d}y}{\mathrm{d}x} + 20y = 6\mathrm{e}^{-12x}$

$\Rightarrow \mathrm{e}^{20x}\frac{\mathrm{d}y}{\mathrm{d}x} + 20\mathrm{e}^{20x}y = 6\mathrm{e}^{-12x} \times \mathrm{e}^{20x}$

$\Rightarrow \mathrm{e}^{20x}\frac{\mathrm{d}y}{\mathrm{d}x} + 20\mathrm{e}^{20x}y = 6\mathrm{e}^{8x}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(\mathrm{e}^{20x}y) = 6\mathrm{e}^{8x}$

$\Rightarrow \mathrm{e}^{20x}y = \frac{3}{4}\mathrm{e}^{8x} + c$

$\Rightarrow y = \frac{3}{4}\mathrm{e}^{-12x} + c\mathrm{e}^{-20x}$

12) a)

$\mathrm{e}^{\ln x} = x$

12) b)

$\mathrm{e}^{\ln x^2} = x^2$

12) c)

$\mathrm{e}^{4\ln x} = \mathrm{e}^{\ln x^4} = x^4$

12) d)

$\mathrm{e}^{-3\ln x} = \mathrm{e}^{\ln x^{-3}} = x^{-3} = \frac{1}{x^3}$

13) a)

Integrating Factor

$= \mathrm{e}^{\int \frac{1}{x} \mathrm{d}x}$

$= \mathrm{e}^{\ln|x|}$

$= |x|$

Let Integrating Factor $= x$

$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y}{x} = x$

$\Rightarrow x\frac{\mathrm{d}y}{\mathrm{d}x} + y = x^2$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(xy) = x^2$

$\Rightarrow xy = \frac{1}{3}x^3 + c$

$\Rightarrow y = \frac{1}{3}x^2 + \frac{c}{x}$

13) b)

Integrating Factor

$= \mathrm{e}^{\int \frac{2}{x} \mathrm{d}x}$

$= \mathrm{e}^{2\ln|x|}$

$= \mathrm{e}^{\ln|x^2|}$

$= x^2$

$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{2y}{x} = 30$

$\Rightarrow x^2\frac{\mathrm{d}y}{\mathrm{d}x} + 2xy = 30x^2$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(x^2y) = 30x^2$

$\Rightarrow x^2y = 10x^3 + c$

$\Rightarrow y = 10x + \frac{c}{x^2}$

13) c)

Integrating Factor

$= \mathrm{e}^{\int -\frac{3}{x}\mathrm{d}x}$

$= \mathrm{e}^{-3\ln|x|}$

$= \mathrm{e}^{\ln|x^{-3}|}$

$= \left|\frac{1}{x^3}\right|$

Let Integrating Factor $= \frac{1}{x^3}$

$\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{3y}{x} = x^3$

$\Rightarrow \frac{1}{x^3}\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{3y}{x^4} = 1$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{y}{x^3}\right) = 1$

$\Rightarrow \frac{y}{x^3} = x + c$

$\Rightarrow y = x^4 + cx^3$

13) d)

Integrating Factor

$= \mathrm{e}^{\int -\frac{5}{x}\mathrm{d}x}$

$= \mathrm{e}^{-5\ln|x|}$

$= \mathrm{e}^{\ln|x^{-5}|}$

$= \left|\frac{1}{x^5}\right|$

Let Integrating Factor $= \frac{1}{x^5}$

$\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{5y}{x} = \frac{40}{x^4}$

$\Rightarrow \frac{1}{x^5}\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{5y}{x^6} = \frac{40}{x^9}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{y}{x^5}\right) = 40x^{-9}$

$\Rightarrow \frac{y}{x^5} = -5x^{-8} + c$

$\Rightarrow y = -\frac{5}{x^3} + cx^5$

14) a)

Integrating Factor

$= \mathrm{e}^{\int \frac{1}{x+1}\mathrm{d}x}$

$= \mathrm{e}^{\ln|x+1|}$

$= |x+1|$

Let Integrating Factor $= x+1$

$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y}{x+1} = 6x$

$\Rightarrow (x+1)\frac{\mathrm{d}y}{\mathrm{d}x} + y = 6x(x+1)$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\big((x+1)y\big) = 6x^2 + 6x$

$\Rightarrow (x+1)y = 2x^3 + 3x^2 + c$

$\Rightarrow y = \frac{2x^3+3x^2+c}{x+1}$

14) b)

Integrating Factor

$= \mathrm{e}^{\int \frac{2}{x-3}\mathrm{d}x}$

$= \mathrm{e}^{2\ln|x-3|}$

$= \mathrm{e}^{\ln|(x-3)^2|}$

$= (x-3)^2$

$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{2y}{x-3} = 12$

$\Rightarrow (x-3)^2\frac{\mathrm{d}y}{\mathrm{d}x} + 2(x-3)y = 12(x-3)^2$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}((x-3)^2 y) = 12(x-3)^2$

$\Rightarrow (x-3)^2 y = 4(x-3)^3 + c$

$\Rightarrow y = 4(x-3) + \frac{c}{(x-3)^2}$

14) c)

Integrating Factor

$= \mathrm{e}^{\int -\frac{4}{2x+1}\mathrm{d}x}$

$= \mathrm{e}^{-2\ln|2x+1|}$

$= \mathrm{e}^{\ln|(2x+1)^{-2}|}$

$= (2x+1)^{-2}$

$\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{4y}{2x+1} = 2$

$\Rightarrow \frac{1}{(2x+1)^2}\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{4y}{(2x+1)^3} = 2(2x+1)^{-2}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{(2x+1)^2}y\right) = 2(2x+1)^{-2}$

$\Rightarrow \frac{1}{(2x+1)^2}y = -(2x+1)^{-1} + c$

$\Rightarrow y = -(2x+1) + c(2x+1)^2$

15) a)

$p\frac{\mathrm{d}y}{\mathrm{d}x} + qy = r$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$:

$p\frac{\mathrm{d}y}{\mathrm{d}x} + qy = r$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} + \frac{qy}{p} = \frac{r}{p}$

Integrating Factor

$= \mathrm{e}^{\int \frac{q}{p}\mathrm{d}x}$

$= \mathrm{e}^{\frac{q}{p}x}$

Let Integrating Factor $= \mathrm{e}^{\frac{q}{p}x}$

$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{qy}{p} = \frac{r}{p}$

$\Rightarrow \mathrm{e}^{\frac{q}{p}x}\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{qy}{p}\mathrm{e}^{\frac{q}{p}x} = \frac{r}{p}\mathrm{e}^{\frac{q}{p}x}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\left(\mathrm{e}^{\frac{q}{p}x}y\right) = \frac{r}{p}\mathrm{e}^{\frac{q}{p}x}$

$\Rightarrow \mathrm{e}^{\frac{q}{p}x}y = \frac{r}{p} \times \frac{p}{q}\mathrm{e}^{\frac{q}{p}x} + c$

$\Rightarrow \mathrm{e}^{\frac{q}{p}x}y = \frac{r}{q}\mathrm{e}^{\frac{q}{p}x} + c$

$\Rightarrow y = \frac{r}{q} + c\mathrm{e}^{-\frac{q}{p}x}$

Substitute $y = \frac{2r}{q}$ when $x = 0$:

$\frac{2r}{q} = \frac{r}{q} + c\mathrm{e}^{0}$

$\Rightarrow \frac{2r}{q} = \frac{r}{q} + c$

$\Rightarrow c = \frac{r}{q}$

$\therefore y = \frac{r}{q} + \frac{r}{q}\mathrm{e}^{-\frac{q}{p}x}$

15) b)

As $x \to \infty$, $\frac{r}{q}\mathrm{e}^{-\frac{q}{p}x} \to 0$ so $y \to \frac{r}{q}$

16) a)

$x\frac{\mathrm{d}y}{\mathrm{d}x} + y = 1$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$:

$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y}{x} = \frac{1}{x}$

Integrating Factor

$= \mathrm{e}^{\int \frac{1}{x}\mathrm{d}x}$

$= \mathrm{e}^{\ln|x|}$

$= |x|$

Let Integrating Factor $= x$

$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y}{x} = \frac{1}{x}$

$\Rightarrow x\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y}{x} \times x = \frac{1}{x} \times x$

$\Rightarrow x\frac{\mathrm{d}y}{\mathrm{d}x} + y = 1$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(xy) = 1$

$\Rightarrow xy = x + c$

$\Rightarrow y = \frac{x+c}{x}$

16) b)

$(x+1)\frac{\mathrm{d}y}{\mathrm{d}x}+y=2$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x}+\mathrm{P}(x)y=\mathrm{Q}(x)$:

$\frac{\mathrm{d}y}{\mathrm{d}x}+\frac{y}{x+1}=\frac{2}{x+1}$

Integrating Factor

$=\mathrm{e}^{\int\frac{1}{x+1}\mathrm{d}x}$

$=\mathrm{e}^{\ln|x+1|}$

$=|x+1|$

Let Integrating Factor $=x+1$

$\frac{\mathrm{d}y}{\mathrm{d}x}+\frac{y}{x+1}=\frac{2}{x+1}$

$\Rightarrow(x+1)\frac{\mathrm{d}y}{\mathrm{d}x}+\frac{y}{x+1}\times(x+1)=\frac{2}{x+1}\times(x+1)$

$\Rightarrow(x+1)\frac{\mathrm{d}y}{\mathrm{d}x}+y=2$

$\Rightarrow\frac{\mathrm{d}}{\mathrm{d}x}\big((x+1)y\big)=2$

$\Rightarrow(x+1)y=2x+c$

$\Rightarrow y=\frac{2x+c}{x+1}$

16) c)

$x\frac{\mathrm{d}y}{\mathrm{d}x} - y = 1$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$:

$\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{x} = \frac{1}{x}$

Integrating Factor

$= \mathrm{e}^{\int -\frac{1}{x}\mathrm{d}x}$

$= \mathrm{e}^{-\ln|x|}$

$= \mathrm{e}^{\ln|x^{-1}|}$

$= |x^{-1}|$

Let Integrating Factor $= x^{-1}$

$\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{x} = \frac{1}{x}$

$\Rightarrow x^{-1}\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{x} \times x^{-1} = \frac{1}{x} \times x^{-1}$

$\Rightarrow x^{-1}\frac{\mathrm{d}y}{\mathrm{d}x} - x^{-2}y = x^{-2}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(x^{-1}y) = x^{-2}$

$\Rightarrow x^{-1}y = -x^{-1} + c$

$\Rightarrow y = -1 + cx$

16) d)

$x\frac{\mathrm{d}y}{\mathrm{d}x} - y = 2x^3$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$:

$\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{x} = 2x^2$

Integrating Factor

$= \mathrm{e}^{\int -\frac{1}{x}\mathrm{d}x}$

$= \mathrm{e}^{-\ln|x|}$

$= \mathrm{e}^{\ln|x^{-1}|}$

$= |x^{-1}|$

Let Integrating Factor $= x^{-1}$

$\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{x} = 2x^2$

$\Rightarrow x^{-1}\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{x} \times x^{-1} = 2x^2 \times x^{-1}$

$\Rightarrow x^{-1}\frac{\mathrm{d}y}{\mathrm{d}x} - x^{-2}y = 2x$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(x^{-1}y) = 2x$

$\Rightarrow x^{-1}y = x^2 + c$

$\Rightarrow y = x^3 + cx$

17) a)

$x\frac{\mathrm{d}y}{\mathrm{d}x} + 3y = \sqrt{x}$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$:

$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{3y}{x} = \frac{\sqrt{x}}{x}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} + \frac{3y}{x} = x^{-\frac{1}{2}}$

Integrating Factor

$= \mathrm{e}^{\int \frac{3}{x}\mathrm{d}x}$

$= \mathrm{e}^{3\ln|x|}$

$= \mathrm{e}^{\ln|x^3|}$

$= |x^3|$

Let Integrating Factor $= x^3$

$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{3y}{x} = x^{-\frac{1}{2}}$

$\Rightarrow x^3\frac{\mathrm{d}y}{\mathrm{d}x} + 3x^2y = x^{\frac{5}{2}}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(x^3y) = x^{\frac{5}{2}}$

$\Rightarrow x^3y = \frac{2}{7}x^{\frac{7}{2}} + c$

$\Rightarrow y = \frac{2}{7}x^{\frac{1}{2}} + cx^{-3}$

17) b)

$x\frac{\mathrm{d}y}{\mathrm{d}x} - y = \frac{1}{\sqrt{x}}$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$:

$\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{x} = \frac{1}{x\sqrt{x}}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{x} = x^{-\frac{3}{2}}$

Integrating Factor

$= \mathrm{e}^{\int -\frac{1}{x}\mathrm{d}x}$

$= \mathrm{e}^{-\ln|x|}$

$= \mathrm{e}^{\ln|x^{-1}|}$

$= \left|\frac{1}{x}\right|$

Let Integrating Factor $= \frac{1}{x}$

$\frac{1}{x}\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{x^2} = x^{-\frac{5}{2}}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{x}y\right) = x^{-\frac{5}{2}}$

$\Rightarrow \frac{1}{x}y = -\frac{2}{3}x^{-\frac{3}{2}} + c$

$\Rightarrow y = -\frac{2}{3}x^{-\frac{1}{2}} + cx$

18) a)

$2x^{\frac{5}{2}}\frac{\mathrm{d}y}{\mathrm{d}x}+x^{\frac{3}{2}}y=2$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x}+\mathrm{P}(x)y=\mathrm{Q}(x)$:

$\frac{\mathrm{d}y}{\mathrm{d}x}+\frac{x^{\frac{3}{2}}y}{2x^{\frac{5}{2}}}=\frac{2}{2x^{\frac{5}{2}}}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x}+\frac{1}{2x}y=x^{-\frac{5}{2}}$

Integrating Factor

$=\mathrm{e}^{\int\frac{1}{2x}\mathrm{d}x}$

$=\mathrm{e}^{\frac{1}{2}\ln|x|}$

$=\mathrm{e}^{\ln\left|x^{\frac{1}{2}}\right|}$

$=\left|x^{\frac{1}{2}}\right|$

$=x^{\frac{1}{2}}$

$\frac{\mathrm{d}y}{\mathrm{d}x}+\frac{1}{2x}y=x^{-\frac{5}{2}}$

$\Rightarrow x^{\frac{1}{2}}\frac{\mathrm{d}y}{\mathrm{d}x}+\frac{1}{2x}\times x^{\frac{1}{2}}\times y=x^{-\frac{5}{2}}\times x^{\frac{1}{2}}$

$\Rightarrow x^{\frac{1}{2}}\frac{\mathrm{d}y}{\mathrm{d}x}+\frac{1}{2}x^{-\frac{1}{2}}y=x^{-2}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\left(x^{\frac{1}{2}}y\right)=x^{-2}$

$\Rightarrow x^{\frac{1}{2}}y=-x^{-1}+c$

$\Rightarrow y=-x^{-\frac{3}{2}}+cx^{-\frac{1}{2}}$

18) b)

$2x^{\frac{5}{2}}\frac{\mathrm{d}y}{\mathrm{d}x} - x^{\frac{3}{2}}y = 2$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$:

$\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{x^{\frac{3}{2}}y}{2x^{\frac{5}{2}}} = \frac{2}{2x^{\frac{5}{2}}}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} - \frac{1}{2x}y = x^{-\frac{5}{2}}$

Integrating Factor

$= \mathrm{e}^{\int -\frac{1}{2x}\mathrm{d}x}$

$= \mathrm{e}^{-\frac{1}{2}\ln|x|}$

$= \mathrm{e}^{\ln\left|x^{-\frac{1}{2}}\right|}$

$= \left|x^{-\frac{1}{2}}\right|$

$= x^{-\frac{1}{2}}$

$\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{1}{2x}y = x^{-\frac{5}{2}}$

$\Rightarrow x^{-\frac{1}{2}}\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{1}{2x} \times x^{-\frac{1}{2}} \times y = x^{-\frac{5}{2}} \times x^{-\frac{1}{2}}$

$\Rightarrow x^{-\frac{1}{2}}\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{1}{2}x^{-\frac{3}{2}}y = x^{-3}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\left(x^{-\frac{1}{2}}y\right) = x^{-3}$

$\Rightarrow x^{-\frac{1}{2}}y = -\frac{1}{2}x^{-2} + c$

$\Rightarrow y = -\frac{1}{2}x^{-\frac{3}{2}} + cx^{\frac{1}{2}}$

19)

$x^2\frac{\mathrm{d}y}{\mathrm{d}x}+7xy=\frac{2}{x^5}$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x}+\mathrm{P}(x)y=\mathrm{Q}(x)$:

$x^2\frac{\mathrm{d}y}{\mathrm{d}x}+7xy=\frac{2}{x^5}$

$\Rightarrow\frac{\mathrm{d}y}{\mathrm{d}x}+\frac{7y}{x}=\frac{2}{x^7}$

Integrating Factor

$=\mathrm{e}^{\int\frac{7}{x}\mathrm{d}x}$

$=\mathrm{e}^{7\ln|x|}$

$=\mathrm{e}^{\ln|x^7|}$

$=|x^7|$

Let Integrating Factor $=x^7$

$x^7\frac{\mathrm{d}y}{\mathrm{d}x}+\frac{7y}{x}\times x^7=\frac{2}{x^7}\times x^7$

$\Rightarrow x^7\frac{\mathrm{d}y}{\mathrm{d}x}+7x^6y=2$

$\Rightarrow\frac{\mathrm{d}}{\mathrm{d}x}(x^7y)=2$

$\Rightarrow x^7y=2x+c$

$\Rightarrow y=\frac{2}{x^6}+\frac{c}{x^7}$

Substitute $y=1$ and $x=2$:

$1=\frac{2}{2^6}+\frac{c}{2^7}$

$\Rightarrow 1=\frac{1}{32}+\frac{c}{128}$

$\Rightarrow c=124$

$\therefore y=\frac{2}{x^6}+\frac{124}{x^7}$

$y=2x^{-6}+124x^{-7}$

$\frac{\mathrm{d}y}{\mathrm{d}x}=-12x^{-7}-868x^{-8}$

As $x>0$, $-12x^{-7}<0$ and $-868x^{-8}<0$

So $\frac{\mathrm{d}y}{\mathrm{d}x}<0$ for all $x>0$, and so y is a decreasing function.

20) a)

$\sin x \frac{\mathrm{d}y}{\mathrm{d}x} - y\cos x = 2^x \sin^2 x$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$:

$\sin x \frac{\mathrm{d}y}{\mathrm{d}x} - y\cos x = 2^x \sin^2 x$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y\cos x}{\sin x} = 2^x \sin x$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} - y\cot x = 2^x \sin x$

Integrating Factor

$= \mathrm{e}^{-\int \cot x \mathrm{d}x}$

$= \mathrm{e}^{-\ln|\sin x|}$

$= \mathrm{e}^{\ln|(\sin x)^{-1}|}$

$= |\mathrm{cosec}\, x|$

Let Integrating Factor $= \mathrm{cosec}\, x$

$\mathrm{cosec}\, x \frac{\mathrm{d}y}{\mathrm{d}x} - y\cot x \,\mathrm{cosec}\, x = 2^x \sin x \,\mathrm{cosec}\, x$

$\Rightarrow \mathrm{cosec}\, x \frac{\mathrm{d}y}{\mathrm{d}x} - y\cot x \,\mathrm{cosec}\, x = 2^x$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(y\,\mathrm{cosec}\, x) = 2^x$

$\Rightarrow y\,\mathrm{cosec}\, x = \frac{1}{\ln 2} 2^x + c$

$\Rightarrow y = \frac{2^x}{\ln 2}\sin x + c\sin x$

20) b)

$0 = \frac{2^{\frac{\pi}{2}}}{\ln 2}\sin\frac{\pi}{2} + c\sin\frac{\pi}{2}$

$\Rightarrow 0 = \frac{2^{\frac{\pi}{2}}}{\ln 2} + c$

$\Rightarrow c = -\frac{2^{\frac{\pi}{2}}}{\ln 2}$

$\therefore y = \frac{2^x}{\ln 2}\sin x - \frac{2^{\frac{\pi}{2}}}{\ln 2}\sin x$

$\Rightarrow y = \frac{\sin x}{\ln 2}\left(2^x - 2^{\frac{\pi}{2}}\right)$

21) a)

Integrating Factor

$= \mathrm{e}^{\int x \mathrm{d}x}$

$= \mathrm{e}^{\frac{1}{2}x^2}$

$\frac{\mathrm{d}y}{\mathrm{d}x} + xy = x$

$\Rightarrow \mathrm{e}^{\frac{1}{2}x^2}\frac{\mathrm{d}y}{\mathrm{d}x} + x\mathrm{e}^{\frac{1}{2}x^2}y = x\mathrm{e}^{\frac{1}{2}x^2}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\left(\mathrm{e}^{\frac{1}{2}x^2}y\right) = x\mathrm{e}^{\frac{1}{2}x^2}$

$\Rightarrow \mathrm{e}^{\frac{1}{2}x^2}y = \mathrm{e}^{\frac{1}{2}x^2} + c$

$\Rightarrow y = 1 + c\mathrm{e}^{-\frac{1}{2}x^2}$

21) b)

Integrating Factor

$= \mathrm{e}^{\int -6x^3 \mathrm{d}x}$

$= \mathrm{e}^{-\frac{3}{2}x^4}$

$\frac{\mathrm{d}y}{\mathrm{d}x} - 6x^3y = 8x^3$

$\Rightarrow \mathrm{e}^{-\frac{3}{2}x^4}\frac{\mathrm{d}y}{\mathrm{d}x} - 6x^3\mathrm{e}^{-\frac{3}{2}x^4}y = 8x^3\mathrm{e}^{-\frac{3}{2}x^4}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\left(\mathrm{e}^{-\frac{3}{2}x^4}y\right) = 8x^3\mathrm{e}^{-\frac{3}{2}x^4}$

$\Rightarrow \mathrm{e}^{-\frac{3}{2}x^4}y = -\frac{4}{3}\mathrm{e}^{-\frac{3}{2}x^4} + c$

$\Rightarrow y = -\frac{4}{3} + c\mathrm{e}^{\frac{3}{2}x^4}$

22) a)

Integrating Factor

$= \mathrm{e}^{\int \cos x \mathrm{d}x}$

$= \mathrm{e}^{\sin x}$

$\frac{\mathrm{d}y}{\mathrm{d}x} + y\cos x = \cos x$

$\Rightarrow \mathrm{e}^{\sin x}\frac{\mathrm{d}y}{\mathrm{d}x} + y\cos x\,\mathrm{e}^{\sin x} = \cos x\,\mathrm{e}^{\sin x}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\left(\mathrm{e}^{\sin x}y\right) = \cos x\,\mathrm{e}^{\sin x}$

$\Rightarrow \mathrm{e}^{\sin x}y = \mathrm{e}^{\sin x} + c$

$\Rightarrow y = 1 + c\mathrm{e}^{-\sin x}$

22) b)

Integrating Factor

$= \mathrm{e}^{\int -\sin 2x \mathrm{d}x}$

$= \mathrm{e}^{\frac{1}{2}\cos 2x}$

$\frac{\mathrm{d}y}{\mathrm{d}x} - y\sin 2x = 4\sin 2x$

$\Rightarrow \mathrm{e}^{\frac{1}{2}\cos 2x}\frac{\mathrm{d}y}{\mathrm{d}x} - y\sin 2x\,\mathrm{e}^{\frac{1}{2}\cos 2x} = 4\sin 2x\,\mathrm{e}^{\frac{1}{2}\cos 2x}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\left(\mathrm{e}^{\frac{1}{2}\cos 2x}y\right) = 4\sin 2x\,\mathrm{e}^{\frac{1}{2}\cos 2x}$

$\Rightarrow \mathrm{e}^{\frac{1}{2}\cos 2x}y = -4\mathrm{e}^{\frac{1}{2}\cos 2x} + c$

$\Rightarrow y = -4 + c\mathrm{e}^{-\frac{1}{2}\cos 2x}$

23) a)

$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{2y}{x} = 6\sqrt{x^3+2}$

Integrating Factor

$= \mathrm{e}^{\int \frac{2}{x}\mathrm{d}x}$

$= \mathrm{e}^{2\ln|x|}$

$= \mathrm{e}^{\ln|x^2|}$

$= |x^2|$

Let Integrating Factor $= x^2$

$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{2y}{x} = 6\sqrt{x^3+2}$

$\Rightarrow x^2\frac{\mathrm{d}y}{\mathrm{d}x} + 2xy = 6x^2\sqrt{x^3+2}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(x^2y) = 6x^2(x^3+2)^{\frac{1}{2}}$

$\Rightarrow x^2y = \frac{4}{3}(x^3+2)^{\frac{3}{2}} + c$

$\Rightarrow y = \frac{4(x^3+2)^{\frac{3}{2}}}{3x^2} + \frac{c}{x^2}$

23) b)

$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{3y}{x} = \frac{6}{\sqrt{x^4+2}}$

Integrating Factor

$= \mathrm{e}^{\int \frac{3}{x}\mathrm{d}x}$

$= \mathrm{e}^{3\ln|x|}$

$= \mathrm{e}^{\ln|x^3|}$

$= |x^3|$

Let Integrating Factor $= x^3$

$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{3y}{x} = \frac{6}{\sqrt{x^4+2}}$

$\Rightarrow x^3\frac{\mathrm{d}y}{\mathrm{d}x} + 3x^2y = \frac{6x^3}{\sqrt{x^4+2}}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(x^3y) = \frac{6x^3}{\sqrt{x^4+2}}$

$\Rightarrow x^3y = 3\sqrt{x^4+2} + c$

$\Rightarrow y = \frac{3\sqrt{x^4+2}}{x^3} + \frac{c}{x^3}$

24)

$\frac{dy}{dx} + \frac{6y}{3x-2} = 54(3x-2)^6$

Integrating Factor

$= e^{\int \frac{6}{3x-2} dx}$

$= e^{2\ln|3x-2|}$

$= e^{\ln|(3x-2)^2|}$

$= |(3x-2)^2|$

Let Integrating Factor $= (3x-2)^2$

$\frac{dy}{dx} + \frac{6y}{3x-2} = 54(3x-2)^6$

$\Rightarrow (3x-2)^2 \frac{dy}{dx} + 6(3x-2)y = 54(3x-2)^8$

$\Rightarrow \frac{d}{dx}((3x-2)^2 y) = 54(3x-2)^8$

$\Rightarrow (3x-2)^2 y = 2(3x-2)^9 + c$

$\Rightarrow y = 2(3x-2)^7 + c(3x-2)^{-2}$

$\frac{dy}{dx} = 42(3x-2)^6 - 6c(3x-2)^{-3}$

Stationary points exist when $\frac{dy}{dx} = 0$

Substitute $\frac{dy}{dx} = 0$ when $x = \frac{1}{3}$:

$0 = 42\left(3 \times \frac{1}{3} - 2\right)^6 - 6c\left(3 \times \frac{1}{3} - 2\right)^{-3}$

$\Rightarrow 0 = 42 + 6c$

$\Rightarrow c = -7$

$\therefore y = 2(3x-2)^7 - 7(3x-2)^{-2}$

25) a)

$\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{x} = 36x^6 \ln x$

Integrating Factor

$= \mathrm{e}^{\int -\frac{1}{x}\mathrm{d}x}$

$= \mathrm{e}^{-\ln|x|}$

$= \mathrm{e}^{\ln|x^{-1}|}$

$= |x^{-1}|$

Let Integrating Factor $= x^{-1} = \frac{1}{x}$

$\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{x} = 36x^6 \ln x$

$\Rightarrow \frac{1}{x}\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{x^2} = 36x^5 \ln x$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{y}{x}\right) = 36x^5 \ln x$

Use Integration by Parts:

$u = \ln x$ $\qquad$ $\frac{\mathrm{d}v}{\mathrm{d}x} = 36x^5$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{x}$ $\qquad$ $v = 6x^6$

$\therefore \int 36x^5 \ln x \,\mathrm{d}x = 6x^6 \ln x - \int 6x^5 \,\mathrm{d}x$

$= 6x^6 \ln x - x^6 + c$

$\therefore \frac{y}{x} = 6x^6 \ln x - x^6 + c$

$\Rightarrow y = 6x^7 \ln x - x^7 + cx$

Substitute $y = 2$ when $x = 1$:

$2 = 6 \times 1^7 \times \ln 1 - 1^7 + c \times 1$

$\Rightarrow 2 = 0 - 1 + c$

$\Rightarrow c = 3$

$\therefore y = 6x^7 \ln x - x^7 + 3x$

25) b)

$x\sqrt{x}\frac{\mathrm{d}y}{\mathrm{d}x} + y\sqrt{x} = 8x\sqrt{x^3+4x}$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)$:

$x\sqrt{x}\frac{\mathrm{d}y}{\mathrm{d}x} + y\sqrt{x} = 8x\sqrt{x^3+4x}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} + \frac{\sqrt{x}}{x\sqrt{x}}y = \frac{8x\sqrt{x^3+4x}}{x\sqrt{x}}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} + \frac{1}{x}y = 8\sqrt{x^2+4}$

Integrating Factor

$= \mathrm{e}^{\int \frac{1}{x}\mathrm{d}x}$

$= \mathrm{e}^{\ln|x|}$

$= |x|$

Let Integrating Factor $= x$

$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{1}{x}y = 8\sqrt{x^2+4}$

$\Rightarrow x\frac{\mathrm{d}y}{\mathrm{d}x} + y = 8x\sqrt{x^2+4}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(xy) = 8x\sqrt{x^2+4}$

$\Rightarrow xy = \frac{8}{3}(x^2+4)^{\frac{3}{2}} + c$

$\Rightarrow y = \frac{8}{3x}(x^2+4)^{\frac{3}{2}} + \frac{c}{x}$

Substitute $y = 1$ when $x = \sqrt{5}$:

$1 = \frac{8}{3\sqrt{5}}\left(\left(\sqrt{5}\right)^2+4\right)^{\frac{3}{2}} + \frac{c}{\sqrt{5}}$

$\Rightarrow 1 = \frac{8}{3\sqrt{5}} \times 27 + \frac{c}{\sqrt{5}}$

$\Rightarrow 1 = \frac{72}{\sqrt{5}} + \frac{c}{\sqrt{5}}$

$\Rightarrow \sqrt{5} = 72 + c$

$\Rightarrow c = \sqrt{5} - 72$

$\therefore y = \frac{8}{3x}(x^2+4)^{\frac{3}{2}} + \frac{\sqrt{5}-72}{x}$

25) c)

$\tan x \frac{\mathrm{d}y}{\mathrm{d}x} + y \sec^2 x = \tan^2 x$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$:

$\tan x \frac{\mathrm{d}y}{\mathrm{d}x} + y \sec^2 x = \tan^2 x$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y \sec^2 x}{\tan x} = \tan x$

Integrating Factor

$= \mathrm{e}^{\int \frac{\sec^2 x}{\tan x} \mathrm{d}x}$

$= \mathrm{e}^{\ln|\tan x|}$

$= |\tan x|$

Let Integrating Factor $= \tan x$

$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y \sec^2 x}{\tan x} = \tan x$

$\Rightarrow \tan x \frac{\mathrm{d}y}{\mathrm{d}x} + y \sec^2 x = \tan^2 x$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(y \tan x) = \tan^2 x$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(y \tan x) = \sec^2 x - 1$

$\Rightarrow y \tan x = \tan x - x + c$

$\Rightarrow y = 1 - \frac{x}{\tan x} + \frac{c}{\tan x}$

Substitute $y = 1$ when $x = \frac{\pi}{4}$:

$1 = 1 - \frac{\frac{\pi}{4}}{\tan\frac{\pi}{4}} + \frac{c}{\tan\frac{\pi}{4}}$

$\Rightarrow 1 = 1 - \frac{\frac{\pi}{4}}{1} + \frac{c}{1}$

$\Rightarrow 1 = 1 - \frac{\pi}{4} + c$

$\Rightarrow c = \frac{\pi}{4}$

$\therefore y = 1 - \frac{x}{\tan x} + \frac{\frac{\pi}{4}}{\tan x}$

$\Rightarrow y = 1 + \frac{\frac{\pi}{4} - x}{\tan x}$

26)

$\frac{\mathrm{d}y}{\mathrm{d}x} = k(y - x)$

The gradient is -8 at $(1, 3)$:

$-8 = k(3 - 1) \Rightarrow k = -4$

$\therefore \frac{\mathrm{d}y}{\mathrm{d}x} = -4(y - x)$

Rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$:

$\frac{\mathrm{d}y}{\mathrm{d}x} = -4y + 4x$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} + 4y = 4x$

Integrating Factor

$= \mathrm{e}^{\int 4\mathrm{d}x}$

$= \mathrm{e}^{4x}$

$\frac{\mathrm{d}y}{\mathrm{d}x} + 4y = 4x$

$\Rightarrow \mathrm{e}^{4x}\frac{\mathrm{d}y}{\mathrm{d}x} + 4\mathrm{e}^{4x}y = 4x\mathrm{e}^{4x}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(y\mathrm{e}^{4x}) = 4x\mathrm{e}^{4x}$

Use Integration by Parts:

$u = 4x$	$\frac{\mathrm{d}v}{\mathrm{d}x} = \mathrm{e}^{4x}$
$\frac{\mathrm{d}u}{\mathrm{d}x} = 4$	$v = \frac{1}{4}\mathrm{e}^{4x}$

$\therefore \int 4x\mathrm{e}^{4x}\,\mathrm{d}x = x\mathrm{e}^{4x} - \int \mathrm{e}^{4x}\,\mathrm{d}x$

$= x\mathrm{e}^{4x} - \frac{1}{4}\mathrm{e}^{4x} + c$

$\Rightarrow y\mathrm{e}^{4x} = x\mathrm{e}^{4x} - \frac{1}{4}\mathrm{e}^{4x} + c$

$\Rightarrow y = x - \frac{1}{4} + c\mathrm{e}^{-4x}$

Substitute $y = 3$ and $x = 1$:

$3 = 1 - \frac{1}{4} + c\mathrm{e}^{-4}$

$\Rightarrow c = \frac{9}{4}\mathrm{e}^{4}$

$\therefore y = x - \frac{1}{4} + \frac{9}{4}\mathrm{e}^{4}\mathrm{e}^{-4x}$

$\Rightarrow y = x - \frac{1}{4} + \frac{9}{4}\mathrm{e}^{4-4x}$

27) a)

$(1-\cos 2x)\frac{\mathrm{d}y}{\mathrm{d}x}+2y\sin 2x=2(\sin x+\cos x)$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x}+\mathrm{P}(x)y=\mathrm{Q}(x)$:

$(1-\cos 2x)\frac{\mathrm{d}y}{\mathrm{d}x}+2y\sin 2x=2(\sin x+\cos x)$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x}+\frac{2\sin 2x}{1-\cos 2x}y=\frac{2(\sin x+\cos x)}{1-\cos 2x}$

Integrating Factor

$=\mathrm{e}^{\int\frac{2\sin 2x}{1-\cos 2x}\mathrm{d}x}$

$=\mathrm{e}^{\ln|1-\cos 2x|}$

$=|1-\cos 2x|$

Let Integrating Factor $=1-\cos 2x$

$\frac{\mathrm{d}y}{\mathrm{d}x}+\frac{2\sin 2x}{1-\cos 2x}y=\frac{2(\sin x+\cos x)}{1-\cos 2x}$

$\Rightarrow (1-\cos 2x)\frac{\mathrm{d}y}{\mathrm{d}x}+\frac{2\sin 2x}{1-\cos 2x}y\times(1-\cos 2x)=\frac{2(\sin x+\cos x)}{1-\cos 2x}\times(1-\cos 2x)$

$\Rightarrow (1-\cos 2x)\frac{\mathrm{d}y}{\mathrm{d}x}+2y\sin 2x=2(\sin x+\cos x)$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\big((1-\cos 2x)y\big)=2(\sin x+\cos x)$

$\Rightarrow (1-\cos 2x)y=2(-\cos x+\sin x)+c$

$\Rightarrow y=\frac{2\sin x-2\cos x+c}{1-\cos 2x}$

27) b)

$5=\frac{2\sin\frac{\pi}{2}-2\cos\frac{\pi}{2}+c}{1-\cos\left(2\times\frac{\pi}{2}\right)}$

$\Rightarrow 5=\frac{2\sin\frac{\pi}{2}-2\cos\frac{\pi}{2}+c}{1-\cos\pi}$

$\Rightarrow 5=\frac{2-0+c}{1-(-1)}$

$\Rightarrow 5=\frac{2+c}{2}$

$\Rightarrow c=8$

$\therefore y=\frac{2\sin x-2\cos x+8}{1-\cos 2x}$

$\Rightarrow y=\frac{2\sin x-2\cos x+8}{1-(1-2\sin^2 x)}$

$\Rightarrow y=\frac{2\sin x-2\cos x+8}{2\sin^2 x}$

$\Rightarrow y=\frac{\sin x-\cos x+4}{\sin^2 x}$

28) a)

$x\frac{\mathrm{d}y}{\mathrm{d}x}+(2x+1)y=x\mathrm{e}^{x}$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x}+\mathrm{P}(x)y=\mathrm{Q}(x)$:

$\frac{\mathrm{d}y}{\mathrm{d}x}+\frac{2x+1}{x}y=\mathrm{e}^{x}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x}+\left(2+\frac{1}{x}\right)y=\mathrm{e}^{x}$

Integrating Factor

$=\mathrm{e}^{\int\left(2+\frac{1}{x}\right)\mathrm{d}x}$

$=\mathrm{e}^{2x+\ln|x|}$

$=\mathrm{e}^{2x}\mathrm{e}^{\ln|x|}$

$=|x|\mathrm{e}^{2x}$

Let Integrating Factor $=x\mathrm{e}^{2x}$

$\frac{\mathrm{d}y}{\mathrm{d}x}+\left(2+\frac{1}{x}\right)y=\mathrm{e}^{x}$

$\Rightarrow x\mathrm{e}^{2x}\frac{\mathrm{d}y}{\mathrm{d}x}+\left(2+\frac{1}{x}\right)x\mathrm{e}^{2x}y=\mathrm{e}^{x}\times x\mathrm{e}^{2x}$

$\Rightarrow x\mathrm{e}^{2x}\frac{\mathrm{d}y}{\mathrm{d}x}+(2x+1)\mathrm{e}^{2x}y=x\mathrm{e}^{3x}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(x\mathrm{e}^{2x}y)=x\mathrm{e}^{3x}$

Use Integration by Parts:

$$u=x \qquad \frac{\mathrm{d}v}{\mathrm{d}x}=\mathrm{e}^{3x}$$

$$\frac{\mathrm{d}u}{\mathrm{d}x}=1 \qquad v=\frac{1}{3}\mathrm{e}^{3x}$$

$\therefore \int x\mathrm{e}^{3x}\,\mathrm{d}x=\frac{1}{3}x\mathrm{e}^{3x}-\int\frac{1}{3}\mathrm{e}^{3x}\,\mathrm{d}x$

$\Rightarrow \int x\mathrm{e}^{3x}\,\mathrm{d}x=\frac{1}{3}x\mathrm{e}^{3x}-\frac{1}{9}\mathrm{e}^{3x}+c$

So:

$x\mathrm{e}^{2x}y=\frac{1}{3}x\mathrm{e}^{3x}-\frac{1}{9}\mathrm{e}^{3x}+c$

$\Rightarrow y=\frac{x\mathrm{e}^{3x}}{3x\mathrm{e}^{2x}}-\frac{\mathrm{e}^{3x}}{9x\mathrm{e}^{2x}}+\frac{c}{x\mathrm{e}^{2x}}$

$\Rightarrow y=\frac{1}{3}\mathrm{e}^{x}-\frac{1}{9x}\mathrm{e}^{x}+\frac{c}{x\mathrm{e}^{2x}}$

$\Rightarrow y=\left(\frac{1}{3}-\frac{1}{9x}\right)\mathrm{e}^{x}+\frac{c}{x\mathrm{e}^{2x}}$

$\Rightarrow y=\frac{3x-1}{9x}\mathrm{e}^{x}+\frac{c}{x\mathrm{e}^{2x}}$

28) b)

$\frac{1}{3}\mathrm{e} = \frac{3\times1-1}{9\times1}\mathrm{e}^1 + \frac{c}{1\times\mathrm{e}^{2\times1}}$

$\Rightarrow \frac{1}{3}\mathrm{e} = \frac{2}{9}\mathrm{e} + \frac{c}{\mathrm{e}^2}$

$\Rightarrow \frac{1}{9}\mathrm{e} = \frac{c}{\mathrm{e}^2}$

$\Rightarrow c = \frac{1}{9}\mathrm{e}^3$

$\therefore y = \frac{3x-1}{9x}\mathrm{e}^x + \frac{\frac{1}{9}\mathrm{e}^3}{x\mathrm{e}^{2x}}$

$\Rightarrow y = \frac{3x-1}{9x}\mathrm{e}^x + \frac{\mathrm{e}^{3-2x}}{9x}$

$\Rightarrow y = \frac{(3x-1)\mathrm{e}^x+\mathrm{e}^{3-2x}}{9x}$

29)

$\frac{\mathrm{d}y}{\mathrm{d}x} - y\tan x = \sin 2x$

Integrating Factor

$= \mathrm{e}^{\int -\tan x \mathrm{d}x}$

$= \mathrm{e}^{-\ln|\sec x|}$

$= \mathrm{e}^{\ln|(\sec x)^{-1}|}$

$= \mathrm{e}^{\ln|\cos x|}$

$= |\cos x|$

Let Integrating Factor $= \cos x$

$\cos x \frac{\mathrm{d}y}{\mathrm{d}x} - y\tan x \times \cos x = \sin 2x \times \cos x$

$\Rightarrow \cos x \frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y\sin x}{\cos x} \times \cos x = 2\sin x \cos x \times \cos x$

$\Rightarrow \cos x \frac{\mathrm{d}y}{\mathrm{d}x} - y\sin x = 2\sin x \cos^2 x$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(y\cos x) = 2\sin x \cos^2 x$

$\Rightarrow y\cos x = -\frac{2}{3}\cos^3 x + c$

$\Rightarrow y = -\frac{2}{3}\cos^2 x + c\sec x$

Substitute $y = \frac{4}{3}$ when $x = 0$:

$\frac{4}{3} = -\frac{2}{3}\cos^2 0 + c\sec 0$

$\Rightarrow \frac{4}{3} = -\frac{2}{3} + c$

$\Rightarrow c = 2$

$\therefore y = -\frac{2}{3}\cos^2 x + 2\sec x$

30) a)

$\sec x \frac{dy}{dx} + y \operatorname{cosec} x = 2 \tan x$

First, rearrange into the form $\frac{dy}{dx} + P(x)y = Q(x)$:

$\frac{1}{\cos x}\frac{dy}{dx} + \frac{y}{\sin x} = \frac{2\sin x}{\cos x}$

$\Rightarrow \frac{dy}{dx} + \frac{y\cos x}{\sin x} = 2\sin x$

Integrating Factor

$= e^{\int \frac{\cos x}{\sin x} dx}$

$= e^{\ln|\sin x|}$

$= |\sin x|$

Let Integrating Factor $= \sin x$

$\sin x \frac{dy}{dx} + y\cos x = 2\sin^2 x$

$\Rightarrow \frac{d}{dx}(y \sin x) = 2\sin^2 x$

$\cos 2A \equiv 1 - 2\sin^2 A \Rightarrow \sin^2 A \equiv \frac{1}{2} - \frac{1}{2}\cos 2A$

$\Rightarrow \frac{d}{dx}(y \sin x) = 2\left(\frac{1}{2} - \frac{1}{2}\cos 2x\right)$

$\Rightarrow \frac{d}{dx}(y \sin x) = 1 - \cos 2x$

$\Rightarrow y \sin x = x - \frac{1}{2}\sin 2x + c$

$\Rightarrow y = \frac{x}{\sin x} - \frac{\sin 2x}{2\sin x} + \frac{c}{\sin x}$

$\Rightarrow y = x \operatorname{cosec} x - \frac{2\sin x \cos x}{2\sin x} + c \operatorname{cosec} x$

$\Rightarrow y = x \operatorname{cosec} x - \cos x + c \operatorname{cosec} x$

30) b)

$\frac{12}{\sqrt{3}} = \frac{\pi}{3} \times \operatorname{cosec}\frac{\pi}{3} - \cos\frac{\pi}{3} + c \operatorname{cosec}\frac{\pi}{3}$

$\Rightarrow 4\sqrt{3} = \frac{2\pi\sqrt{3}}{9} - \frac{1}{2} + \frac{2c\sqrt{3}}{3}$

$\Rightarrow \frac{2c\sqrt{3}}{3} = 4\sqrt{3} - \frac{2\pi\sqrt{3}}{9} + \frac{1}{2}$

$\Rightarrow c = 6 - \frac{\pi}{3} + \frac{\sqrt{3}}{4}$

$\therefore y = x \operatorname{cosec} x - \cos x + \left(6 - \frac{\pi}{3} + \frac{\sqrt{3}}{4}\right)\operatorname{cosec} x$

31)

$x^4\frac{\mathrm{d}y}{\mathrm{d}x}+\frac{x^3y}{\ln x}=3$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x}+\mathrm{P}(x)y=\mathrm{Q}(x)$:

$x^4\frac{\mathrm{d}y}{\mathrm{d}x}+\frac{x^3y}{\ln x}=3$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x}+\frac{y}{x\ln x}=\frac{3}{x^4}$

Integrating Factor

$=\mathrm{e}^{\int\frac{1}{x\ln x}\mathrm{d}x}$

$=\mathrm{e}^{\ln|\ln x|}$

$=|\ln x|$

Let Integrating Factor $=\ln x$

$\frac{\mathrm{d}y}{\mathrm{d}x}+\frac{y}{x\ln x}=\frac{3}{x^4}$

$\Rightarrow \ln x\frac{\mathrm{d}y}{\mathrm{d}x}+\frac{y}{x\ln x}\times\ln x=\frac{3}{x^4}\times\ln x$

$\Rightarrow \ln x\frac{\mathrm{d}y}{\mathrm{d}x}+\frac{y}{x}=\frac{3\ln x}{x^4}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(y\ln x)=3x^{-4}\ln x$

Use Integration by Parts:

$u=\ln x \qquad \frac{\mathrm{d}v}{\mathrm{d}x}=3x^{-4}$

$\frac{\mathrm{d}u}{\mathrm{d}x}=\frac{1}{x} \qquad v=-x^{-3}$

$\therefore \int 3x^{-4}\ln x\,\mathrm{d}x=-x^{-3}\ln x+\int x^{-4}\,\mathrm{d}x$

$=-x^{-3}\ln x-\frac{1}{3}x^{-3}+c$

$\Rightarrow y\ln x=-x^{-3}\ln x-\frac{1}{3}x^{-3}+c$

$\Rightarrow y=-x^{-3}-\frac{\frac{1}{3}x^{-3}}{\ln x}+\frac{c}{\ln x}$

Substitute $y=\mathrm{e}^{-3}$ when $x=\mathrm{e}$:

$\mathrm{e}^{-3}=-\mathrm{e}^{-3}-\frac{\frac{1}{3}\mathrm{e}^{-3}}{\ln \mathrm{e}}+\frac{c}{\ln \mathrm{e}}$

$\Rightarrow \mathrm{e}^{-3}=-\mathrm{e}^{-3}-\frac{\frac{1}{3}\mathrm{e}^{-3}}{1}+\frac{c}{1}$

$\Rightarrow \mathrm{e}^{-3}=-\mathrm{e}^{-3}-\frac{1}{3}\mathrm{e}^{-3}+c$

$\Rightarrow c=\frac{7}{3}\mathrm{e}^{-3}$

$$\therefore y = -x^{-3} - \frac{\frac{1}{3}x^{-3}}{\ln x} + \frac{\frac{7}{3}e^{-3}}{\ln x}$$
$$\Rightarrow y = -\frac{1}{x^3} - \frac{1}{3x^3 \ln x} + \frac{7e^{-3}}{3\ln x}$$
$$\Rightarrow y = -\frac{3\ln x}{3x^3 \ln x} - \frac{1}{3x^3 \ln x} + \frac{7e^{-3}x^3}{3x^3 \ln x}$$
$$\Rightarrow y = \frac{7e^{-3}x^3 - 3\ln x - 1}{3x^3 \ln x}$$

32) a)

Integrating Factor

$$= e^{\int \sinh x \, dx}$$
$$= e^{\cosh x}$$
$$\frac{dy}{dx} + y\sinh x = 2\sinh x$$
$$\Rightarrow e^{\cosh x}\frac{dy}{dx} + y\sinh x\, e^{\cosh x} = 2\sinh x\, e^{\cosh x}$$
$$\Rightarrow \frac{d}{dx}\left(e^{\cosh x}y\right) = 2\sinh x\, e^{\cosh x}$$
$$\Rightarrow e^{\cosh x}y = 2e^{\cosh x} + c$$
$$\Rightarrow y = 2 + ce^{-\cosh x}$$

32) b)

Integrating Factor

$$= e^{\int -2\cosh 3x \, dx}$$
$$= e^{-\frac{2}{3}\sinh 3x}$$
$$\frac{dy}{dx} - 2y\cosh 3x = 6\cosh 3x$$
$$\Rightarrow e^{-\frac{2}{3}\sinh 3x}\frac{dy}{dx} - 2y\cosh 3x\, e^{-\frac{2}{3}\sinh 3x} = 6\cosh 3x\, e^{-\frac{2}{3}\sinh 3x}$$
$$\Rightarrow \frac{d}{dx}\left(e^{-\frac{2}{3}\sinh 3x}y\right) = 6\cosh 3x\, e^{-\frac{2}{3}\sinh 3x}$$
$$\Rightarrow e^{-\frac{2}{3}\sinh 3x}y = -3e^{-\frac{2}{3}\sinh 3x} + c$$
$$\Rightarrow y = -3 + ce^{\frac{2}{3}\sinh 3x}$$

32) c)

$2\cosh^2 x\frac{\mathrm{d}y}{\mathrm{d}x}+y\sinh 2x=\sinh^5 x\cosh^2 x$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x}+\mathrm{P}(x)y=\mathrm{Q}(x)$:

$\frac{\mathrm{d}y}{\mathrm{d}x}+\frac{y\sinh 2x}{2\cosh^2 x}=\frac{\sinh^5 x\cosh^2 x}{2\cosh^2 x}$

$\Rightarrow\frac{\mathrm{d}y}{\mathrm{d}x}+\frac{2y\sinh x\cosh x}{2\cosh^2 x}=\frac{1}{2}\sinh^5 x$

$\Rightarrow\frac{\mathrm{d}y}{\mathrm{d}x}+y\tanh x=\frac{1}{2}\sinh^5 x$

Integrating Factor

$=\mathrm{e}^{\int\tanh x\mathrm{d}x}$

$=\mathrm{e}^{\ln|\cosh x|}$

$=|\cosh x|$

Let Integrating Factor $=\cosh x$

$\frac{\mathrm{d}y}{\mathrm{d}x}+y\tanh x=\frac{1}{2}\sinh^5 x$

$\Rightarrow\cosh x\frac{\mathrm{d}y}{\mathrm{d}x}+y\tanh x\cosh x=\frac{1}{2}\sinh^5 x\cosh x$

$\Rightarrow\cosh x\frac{\mathrm{d}y}{\mathrm{d}x}+y\sinh x=\frac{1}{2}\sinh^5 x\cosh x$

$\Rightarrow\frac{\mathrm{d}}{\mathrm{d}x}(y\cosh x)=\frac{1}{2}\sinh^5 x\cosh x$

$\Rightarrow y\cosh x=\frac{1}{12}\sinh^6 x+k$

$\Rightarrow y=\frac{\frac{1}{12}\sinh^6 x+k}{\cosh x}$

$\Rightarrow y=\frac{\sinh^6 x+c}{12\cosh x}$

where $c=12k$

33)

$x\frac{\mathrm{d}y}{\mathrm{d}x}+2y=\frac{7-x}{(x^2+7)(x+1)}$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x}+\mathrm{P}(x)y=\mathrm{Q}(x)$:

$x\frac{\mathrm{d}y}{\mathrm{d}x}+2y=\frac{7-x}{(x^2+7)(x+1)}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x}+\frac{2y}{x}=\frac{7-x}{x(x^2+7)(x+1)}$

Integrating Factor

$=\mathrm{e}^{\int\frac{2}{x}\mathrm{d}x}$

$=\mathrm{e}^{2\ln|x|}$

$=\mathrm{e}^{\ln|x^2|}$

$=x^2$

$x^2\frac{\mathrm{d}y}{\mathrm{d}x}+2xy=\frac{7x-x^2}{(x^2+7)(x+1)}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(x^2y)=\frac{7x-x^2}{(x^2+7)(x+1)}$

Use Partial Fractions:

$\frac{7x-x^2}{(x^2+7)(x+1)}\equiv\frac{Ax+B}{x^2+7}+\frac{C}{x+1}$

$\Rightarrow 7x-x^2\equiv(Ax+B)(x+1)+C(x^2+7)$

Let $x=-1$: $-8=8C\Rightarrow C=-1$

Let $x=0$: $0=B-7\Rightarrow B=7$

Let $x=1$: $6=(A+7)\times 2-8\Rightarrow A=0$

$\therefore \frac{7x-x^2}{(x^2+7)(x+1)}\equiv\frac{7}{x^2+7}-\frac{1}{x+1}$

So:

$\frac{\mathrm{d}}{\mathrm{d}x}(x^2y)=\frac{7}{x^2+7}-\frac{1}{x+1}$

$\frac{\mathrm{d}}{\mathrm{d}x}(x^2y)=\frac{7}{7\left(1+\frac{1}{7}x^2\right)}-\frac{1}{x+1}$

$\frac{\mathrm{d}}{\mathrm{d}x}(x^2y)=\frac{1}{1+\left(\frac{1}{\sqrt{7}}x\right)^2}-\frac{1}{x+1}$

$\Rightarrow x^2y=\frac{1}{\frac{1}{\sqrt{7}}}\arctan\left(\frac{1}{\sqrt{7}}x\right)-\ln|x+1|+c$

$\Rightarrow x^2y=\sqrt{7}\arctan\left(\frac{1}{\sqrt{7}}x\right)-\ln|x+1|+c$

$\Rightarrow y=\frac{\sqrt{7}}{x^2}\arctan\left(\frac{x}{\sqrt{7}}\right)-\frac{\ln|x+1|}{x^2}+\frac{c}{x^2}$

34) a)

$e^{\int \frac{1}{1-x^2} dx}$

$= e^{\operatorname{artanh} x}$

$= e^{\frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)}$

$= e^{\ln\left(\left(\frac{1+x}{1-x}\right)^{\frac{1}{2}}\right)}$

$= \left(\frac{1+x}{1-x}\right)^{\frac{1}{2}}$ as required

34) b)

$(1-x^2)\frac{dy}{dx} + y = (1-x)^{\frac{3}{2}}(1+x)$

First, rearrange into the form $\frac{dy}{dx} + P(x)y = Q(x)$:

$(1-x^2)\frac{dy}{dx} + y = (1-x)^{\frac{3}{2}}(1+x)$

$\Rightarrow \frac{dy}{dx} + \frac{y}{1-x^2} = \frac{(1-x)^{\frac{3}{2}}(1+x)}{1-x^2}$

$\Rightarrow \frac{dy}{dx} + \frac{y}{1-x^2} = \frac{(1-x)^{\frac{3}{2}}(1+x)}{(1-x)(1+x)}$

$\Rightarrow \frac{dy}{dx} + \frac{y}{1-x^2} = (1-x)^{\frac{1}{2}}$

Integrating Factor

$= e^{\int \frac{1}{1-x^2} dx}$

$= \left(\frac{1+x}{1-x}\right)^{\frac{1}{2}}$

$\frac{dy}{dx} + \frac{y}{1-x^2} = (1-x)^{\frac{1}{2}}$

$\Rightarrow \left(\frac{1+x}{1-x}\right)^{\frac{1}{2}}\frac{dy}{dx} + \frac{y}{1-x^2}\left(\frac{1+x}{1-x}\right)^{\frac{1}{2}} = (1-x)^{\frac{1}{2}}\left(\frac{1+x}{1-x}\right)^{\frac{1}{2}}$

$\Rightarrow \left(\frac{1+x}{1-x}\right)^{\frac{1}{2}}\frac{dy}{dx} + \frac{y}{1-x^2}\left(\frac{1+x}{1-x}\right)^{\frac{1}{2}} = (1-x)^{\frac{1}{2}} \times \frac{(1+x)^{\frac{1}{2}}}{(1-x)^{\frac{1}{2}}}$

$\Rightarrow \left(\frac{1+x}{1-x}\right)^{\frac{1}{2}}\frac{dy}{dx} + \frac{y}{1-x^2}\left(\frac{1+x}{1-x}\right)^{\frac{1}{2}} = (1+x)^{\frac{1}{2}}$

$\Rightarrow \frac{d}{dx}\left(\left(\frac{1+x}{1-x}\right)^{\frac{1}{2}} y\right) = (1+x)^{\frac{1}{2}}$

$\Rightarrow \left(\frac{1+x}{1-x}\right)^{\frac{1}{2}} y = \frac{2}{3}(1+x)^{\frac{3}{2}} + c$

$\Rightarrow \frac{(1+x)^{\frac{1}{2}}}{(1-x)^{\frac{1}{2}}}y = \frac{2}{3}(1+x)^{\frac{3}{2}} + c$

$\Rightarrow y = \frac{2}{3}(1+x)(1-x)^{\frac{1}{2}} + \frac{c(1-x)^{\frac{1}{2}}}{(1+x)^{\frac{1}{2}}}$

Substitute $y = \frac{5}{3}$ when $x = 0$:

$\frac{5}{3} = \frac{2}{3}(1+0)(1-0)^{\frac{1}{2}} + \frac{c(1-0)^{\frac{1}{2}}}{(1+0)^{\frac{1}{2}}}$

$\Rightarrow \frac{5}{3} = \frac{2}{3} + c$

$\Rightarrow c = 1$

$\therefore y = \frac{2}{3}(1+x)(1-x)^{\frac{1}{2}} + \left(\frac{1-x}{1+x}\right)^{\frac{1}{2}}$

35)

$(9-4x^2)\frac{\mathrm{d}y}{\mathrm{d}x} - 4xy = 2$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$:

$(9-4x^2)\frac{\mathrm{d}y}{\mathrm{d}x} - 4xy = 2$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} - \frac{4x}{9-4x^2}y = \frac{2}{9-4x^2}$

Integrating Factor

$= \mathrm{e}^{\int -\frac{4x}{9-4x^2}\mathrm{d}x}$

$= \mathrm{e}^{\frac{1}{2}\ln|9-4x^2|}$

$= \mathrm{e}^{\ln\left|(9-4x^2)^{\frac{1}{2}}\right|}$

$= \left|\sqrt{9-4x^2}\right|$

Let Integrating Factor $= \sqrt{9-4x^2}$

$\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{4x}{9-4x^2}y = \frac{2}{9-4x^2}$

$\Rightarrow \sqrt{9-4x^2}\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{4x}{9-4x^2}y \times \sqrt{9-4x^2} = \frac{2}{9-4x^2} \times \sqrt{9-4x^2}$

$\Rightarrow \sqrt{9-4x^2}\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{4x}{\sqrt{9-4x^2}}y = \frac{2}{\sqrt{9-4x^2}}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\left(y\sqrt{9-4x^2}\right) = \frac{2}{\sqrt{9-4x^2}}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\left(y\sqrt{9-4x^2}\right) = \frac{2}{\sqrt{9\left(1-\frac{4}{9}x^2\right)}}$

$$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\left(y\sqrt{9-4x^2}\right) = \frac{2}{3\sqrt{1-\left(\frac{2}{3}x\right)^2}}$$

$$\Rightarrow y\sqrt{9-4x^2} = \frac{2}{3} \times \frac{1}{\frac{2}{3}} \times \arcsin\left(\frac{2}{3}x\right) + c$$

$$\Rightarrow y\sqrt{9-4x^2} = \arcsin\left(\frac{2}{3}x\right) + c$$

$$\Rightarrow y = \frac{\arcsin\left(\frac{2}{3}x\right)+c}{\sqrt{9-4x^2}}$$

Substitute $y = 0$ when $x = \frac{3\sqrt{2}}{4}$:

$$0 = \frac{\arcsin\left(\frac{2}{3}\times\frac{3\sqrt{2}}{4}\right)+c}{\sqrt{9-4\left(\frac{3\sqrt{2}}{4}\right)^2}}$$

$$\Rightarrow 0 = \frac{\arcsin\left(\frac{\sqrt{2}}{2}\right)+c}{\sqrt{\frac{9}{2}}}$$

$$\Rightarrow 0 = \frac{\pi}{4} + c$$

$$\Rightarrow c = -\frac{\pi}{4}$$

$$\therefore y = \frac{\arcsin\left(\frac{2}{3}x\right)-\frac{\pi}{4}}{\sqrt{9-4x^2}}$$

36)

$$2x\frac{\mathrm{d}y}{\mathrm{d}x} - y = \sqrt{\frac{4x^3}{4x-x^2-1}}$$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$:

$$2x\frac{\mathrm{d}y}{\mathrm{d}x} - y = \sqrt{\frac{4x^3}{4x-x^2-1}}$$

$$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{2x} = \frac{1}{2x}\sqrt{\frac{4x^3}{4x-x^2-1}}$$

$$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{2x} = \frac{1}{2x}\sqrt{\frac{4x^2\times x}{4x-x^2-1}}$$

$$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{2x} = \sqrt{\frac{x}{4x-x^2-1}}$$

Integrating Factor

$$= \mathrm{e}^{\int -\frac{1}{2x}\mathrm{d}x}$$

$$= \mathrm{e}^{-\frac{1}{2}\int \frac{1}{x}\mathrm{d}x}$$

$$= \mathrm{e}^{-\frac{1}{2}\ln|x|}$$

$= \mathrm{e}^{\ln\left|x^{-\frac{1}{2}}\right|}$

$= \left|x^{-\frac{1}{2}}\right|$

Let Integrating Factor $= x^{-\frac{1}{2}} = \frac{1}{\sqrt{x}}$

$\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{2x} = \sqrt{\frac{x}{4x-x^2-1}}$

$\Rightarrow \frac{1}{\sqrt{x}}\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{2x} \times \frac{1}{\sqrt{x}} = \frac{1}{\sqrt{x}} \times \sqrt{\frac{x}{4x-x^2-1}}$

$\Rightarrow \frac{1}{\sqrt{x}}\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{2x^{\frac{3}{2}}} = \frac{1}{\sqrt{x}} \times \frac{\sqrt{x}}{\sqrt{4x-x^2-1}}$

$\Rightarrow \frac{1}{\sqrt{x}}\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{2x^{\frac{3}{2}}} = \frac{1}{\sqrt{4x-x^2-1}}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{y}{\sqrt{x}}\right) = \frac{1}{\sqrt{4x-x^2-1}}$

Now, $4x - x^2 - 1 \equiv -(x^2 - 4x) - 1 \equiv -((x-2)^2 - 4) - 1 \equiv -(x-2)^2 + 4 - 1$

$\equiv 3 - (x-2)^2$

So:

$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{y}{\sqrt{x}}\right) = \frac{1}{\sqrt{3-(x-2)^2}}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{y}{\sqrt{x}}\right) = \frac{1}{\sqrt{3\left(1-\frac{1}{3}(x-2)^2\right)}}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{y}{\sqrt{x}}\right) = \frac{1}{\sqrt{3}} \times \frac{1}{\sqrt{1-\left(\frac{x-2}{\sqrt{3}}\right)^2}}$

$\Rightarrow \frac{y}{\sqrt{x}} = \frac{1}{\sqrt{3}} \times \frac{1}{\frac{1}{\sqrt{3}}} \times \arcsin\left(\frac{x-2}{\sqrt{3}}\right) + c$

$\Rightarrow y = \sqrt{x}\arcsin\left(\frac{x-2}{\sqrt{3}}\right) + c\sqrt{x}$

Substitute $y = \sqrt{\frac{7}{2}}$ when $x = \frac{7}{2}$:

$\sqrt{\frac{7}{2}} = \sqrt{\frac{7}{2}}\arcsin\left(\frac{\frac{7}{2}-2}{\sqrt{3}}\right) + c\sqrt{\frac{7}{2}}$

$\Rightarrow 1 = \arcsin\frac{\sqrt{3}}{2} + c$

$\Rightarrow 1 = \frac{\pi}{3} + c$

$\Rightarrow c = 1 - \frac{\pi}{3}$

$\therefore y = \sqrt{x}\arcsin\left(\frac{x-2}{\sqrt{3}}\right) + \left(1 - \frac{\pi}{3}\right)\sqrt{x}$

37) a)

Begin with the homogeneous DE: $\frac{\mathrm{d}y}{\mathrm{d}x} + 6y = 0$

Assume $y = \mathrm{e}^{rx}$ so $\frac{\mathrm{d}y}{\mathrm{d}x} = r\mathrm{e}^{rx}$

Then $\frac{\mathrm{d}y}{\mathrm{d}x} + 6y = 0$ becomes:

$r\mathrm{e}^{rx} + 6\mathrm{e}^{rx} = 0$

$\Rightarrow r + 6 = 0$

$\Rightarrow r = -6$

$\therefore y = c\mathrm{e}^{-6x}$

Assume a particular integral of the form $y = A\mathrm{e}^{5x} \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = 5A\mathrm{e}^{5x}$

Then substituting into $\frac{\mathrm{d}y}{\mathrm{d}x} + 6y = \mathrm{e}^{5x}$ we get:

$5A\mathrm{e}^{5x} + 6A\mathrm{e}^{5x} = \mathrm{e}^{5x} \Rightarrow 5A + 6A = 1 \Rightarrow 11A = 1 \Rightarrow A = \frac{1}{11}$

$\therefore$ The General solution is

$y = c\mathrm{e}^{-6x} + \frac{1}{11}\mathrm{e}^{5x}$

37) b)

Begin with the homogeneous DE: $\frac{\mathrm{d}y}{\mathrm{d}x} - 3y = 0$

Assume $y = \mathrm{e}^{rx}$ so $\frac{\mathrm{d}y}{\mathrm{d}x} = r\mathrm{e}^{rx}$

Then $\frac{\mathrm{d}y}{\mathrm{d}x} - 3y = 0$ becomes:

$r\mathrm{e}^{rx} - 3\mathrm{e}^{rx} = 0$

$\Rightarrow r - 3 = 0$

$\Rightarrow r = 3$

$\therefore y = c\mathrm{e}^{3x}$

Assume a particular integral of the form $y = Ax + B \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = A$

Then substituting into $\frac{\mathrm{d}y}{\mathrm{d}x} - 3y = x$ we get:

$A - 3(Ax + B) = x \Rightarrow -3Ax + (A - 3B) = x$

Comparing coefficients: $-3A = 1 \Rightarrow A = -\frac{1}{3}$ and $-\frac{1}{3} - 3B = 0 \Rightarrow B = -\frac{1}{9}$

$\therefore$ The General solution is

$y = c\mathrm{e}^{3x} - \frac{1}{3}x - \frac{1}{9}$

37) c)

Begin with the homogeneous DE: $\frac{\mathrm{d}y}{\mathrm{d}x} + 4y = 0$

Assume $y = \mathrm{e}^{rx}$ so $\frac{\mathrm{d}y}{\mathrm{d}x} = r\mathrm{e}^{rx}$

Then $\frac{\mathrm{d}y}{\mathrm{d}x} + 4y = 0$ becomes:

$r\mathrm{e}^{rx} + 4\mathrm{e}^{rx} = 0$

$\Rightarrow r + 4 = 0$

$\Rightarrow r = -4$

$\therefore y = c\mathrm{e}^{-4x}$

Assume a particular integral of the form $y = A\sin x + B\cos x$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = A\cos x - B\sin x$

Then substituting into $\frac{\mathrm{d}y}{\mathrm{d}x} + 4y = 17\sin x$ we get:

$A\cos x - B\sin x + 4(A\sin x + B\cos x) = 17\sin x$

$\Rightarrow (4A - B)\sin x + (A + 4B)\cos x = 17\sin x$

Comparing coefficients:

$4A - B = 17$ (1)

$A + 4B = 0$ (2)

So $A = 4$ and $B = -1$

$\therefore$ The General solution is

$y = c\mathrm{e}^{-4x} + 4\sin x - \cos x$

37) d)

Begin with the homogeneous DE: $\frac{\mathrm{d}y}{\mathrm{d}x} - 7y = 0$

Assume $y = \mathrm{e}^{rx}$ so $\frac{\mathrm{d}y}{\mathrm{d}x} = r\mathrm{e}^{rx}$

Then $\frac{\mathrm{d}y}{\mathrm{d}x} - 7y = 0$ becomes:

$r\mathrm{e}^{rx} - 7\mathrm{e}^{rx} = 0$

$\Rightarrow r - 7 = 0$

$\Rightarrow r = 7$

$\therefore y = c\mathrm{e}^{7x}$

Assume a particular integral of the form $y = Ax\mathrm{e}^{7x} \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = 7Ax\mathrm{e}^{7x} + A\mathrm{e}^{7x}$

Then substituting into $\frac{\mathrm{d}y}{\mathrm{d}x} - 7y = \mathrm{e}^{7x}$ we get:

$7Ax\mathrm{e}^{7x} + A\mathrm{e}^{7x} - 7Ax\mathrm{e}^{7x} = \mathrm{e}^{7x} \Rightarrow A\mathrm{e}^{7x} = \mathrm{e}^{7x} \Rightarrow A = 1$

$\therefore$ The General solution is

$y = c\mathrm{e}^{7x} + x\mathrm{e}^{7x}$

38) a)

Assume a particular integral of the form $y = Ax + B + C\sin x + D\cos x$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = A + C\cos x - D\sin x$

Then substituting into $\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = 15\cos x - 4x$ we get:

$A + C\cos x - D\sin x + 2(Ax + B + C\sin x + D\cos x) = 15\cos x - 4x$

$\Rightarrow 2Ax + (A + 2B) + (C + 2D)\cos x + (2C - D)\sin x = -4x + 15\cos x$

$2A = -4 \Rightarrow A = -2$

$-2 + 2B = 0 \Rightarrow B = 1$

$C + 2D = 15$ (1)

$2C - D = 0$ (2)

$C = 3$ and $D = 6$

$\therefore$ A particular integral is $y = -2x + 1 + 3\sin x + 6\cos x$

38) b)

Begin with the homogeneous DE: $\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = 0$

Assume $y = \mathrm{e}^{rx}$ so $\frac{\mathrm{d}y}{\mathrm{d}x} = r\mathrm{e}^{rx}$

Then $\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = 0$ becomes:

$r\mathrm{e}^{rx} + 2\mathrm{e}^{rx} = 0$

$\Rightarrow r + 2 = 0$

$\Rightarrow r = -2$

$\therefore y = c\mathrm{e}^{-2x}$

So the General solution is

$y = c\mathrm{e}^{-2x} - 2x + 1 + 3\sin x + 6\cos x$

39)

$\frac{dy}{dx} + y = xy^2$

Let $u = \frac{1}{y} \Rightarrow \frac{du}{dx} = -\frac{1}{y^2}\frac{dy}{dx} \Rightarrow \frac{dy}{dx} = -y^2\frac{du}{dx}$

$u = \frac{1}{y} \Rightarrow y = \frac{1}{u}$

$\therefore -y^2\frac{du}{dx} + y = xy^2$

$\Rightarrow -\frac{1}{u^2}\frac{du}{dx} + \frac{1}{u} = \frac{x}{u^2}$

Rearrange into the form $\frac{du}{dx} + P(x)u = Q(x)$:

$\Rightarrow \frac{du}{dx} - u = -x$

Integrating Factor

$= e^{\int -1 dx}$

$= e^{-x}$

$\frac{du}{dx} - u = -x$

$\Rightarrow e^{-x}\frac{du}{dx} - ue^{-x} = -xe^{-x}$

$\Rightarrow \frac{d}{dx}(ue^{-x}) = -xe^{-x}$

Use Integration by Parts:

$u = -x$	$\frac{dv}{dx} = e^{-x}$
$\frac{du}{dx} = -1$	$v = -e^{-x}$

$\therefore \int -xe^{-x}\,dx = xe^{-x} - \int e^{-x}\,dx$

$= xe^{-x} + e^{-x} + c$

$\Rightarrow ue^{-x} = xe^{-x} + e^{-x} + c$

$\Rightarrow u = x + 1 + ce^{x}$

$\Rightarrow \frac{1}{y} = x + 1 + ce^{x}$

$\Rightarrow y = \frac{1}{x+1+ce^{x}}$

40)

$5xy^4\frac{dy}{dx}+3y^5=\frac{\sin 2x}{x^2}$

Let $u=y^5 \Rightarrow \frac{du}{dx}=5y^4\frac{dy}{dx} \Rightarrow \frac{dy}{dx}=\frac{1}{5y^4}\frac{du}{dx}$

$\therefore 5xy^4\times\frac{1}{5y^4}\frac{du}{dx}+3u=\frac{\sin 2x}{x^2}$

$\Rightarrow x\frac{du}{dx}+3u=\frac{\sin 2x}{x^2}$

Rearrange into the form $\frac{du}{dx}+P(x)u=Q(x)$:

$x\frac{du}{dx}+3u=\frac{\sin 2x}{x^2}$

$\Rightarrow \frac{du}{dx}+\frac{3u}{x}=\frac{\sin 2x}{x^3}$

Integrating Factor

$=e^{\int\frac{3}{x}dx}$

$=e^{3\ln|x|}$

$=e^{\ln|x^3|}$

$=|x^3|$

Let Integrating Factor $=x^3$

$\frac{du}{dx}+\frac{3u}{x}=\frac{\sin 2x}{x^3}$

$\Rightarrow x^3\frac{du}{dx}+3x^2u=\sin 2x$

$\Rightarrow \frac{d}{dx}(x^3u)=\sin 2x$

$\Rightarrow x^3u=-\frac{1}{2}\cos 2x+k$

$\Rightarrow u=\frac{-\frac{1}{2}\cos 2x+k}{x^3}$

$\Rightarrow y^5=\frac{k-\frac{1}{2}\cos 2x}{x^3}$

$\Rightarrow y=\sqrt[5]{\frac{k-\frac{1}{2}\cos 2x}{x^3}}$

$\Rightarrow y=\sqrt[5]{\frac{c-\cos 2x}{2x^3}}$

where $c=2k$

41)

$\frac{\mathrm{d}y}{\mathrm{d}x} + 4xy^3 = y$

Let $u = \frac{1}{y^2} \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = -\frac{2}{y^3}\frac{\mathrm{d}y}{\mathrm{d}x} \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{1}{2}y^3\frac{\mathrm{d}u}{\mathrm{d}x}$

$\therefore -\frac{1}{2}y^3\frac{\mathrm{d}u}{\mathrm{d}x} + 4xy^3 = y$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} - 8x = -\frac{2}{y^2}$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} - 8x = -2u$

Rearrange into the form $\frac{\mathrm{d}u}{\mathrm{d}x} + P(x)u = Q(x)$:

$\frac{\mathrm{d}u}{\mathrm{d}x} + 2u = 8x$

Integrating Factor

$= \mathrm{e}^{\int 2\mathrm{d}x}$

$= \mathrm{e}^{2x}$

$\frac{\mathrm{d}u}{\mathrm{d}x} + 2u = 8x$

$\Rightarrow \mathrm{e}^{2x}\frac{\mathrm{d}u}{\mathrm{d}x} + 2\mathrm{e}^{2x}u = 8x\mathrm{e}^{2x}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(\mathrm{e}^{2x}u) = 8x\mathrm{e}^{2x}$

Use Integration by Parts:

$u = 8x$	$\frac{\mathrm{d}v}{\mathrm{d}x} = \mathrm{e}^{2x}$
$\frac{\mathrm{d}u}{\mathrm{d}x} = 8$	$v = \frac{1}{2}\mathrm{e}^{2x}$

$\therefore \int 8x\mathrm{e}^{2x}\,\mathrm{d}x = 4x\mathrm{e}^{2x} - \int 4\mathrm{e}^{2x}\,\mathrm{d}x$

$= 4x\mathrm{e}^{2x} - 2\mathrm{e}^{2x} + c$

$\Rightarrow \mathrm{e}^{2x}u = 4x\mathrm{e}^{2x} - 2\mathrm{e}^{2x} + c$

$\Rightarrow u = 4x - 2 + c\mathrm{e}^{-2x}$

$\Rightarrow \frac{1}{y^2} = 4x - 2 + c\mathrm{e}^{-2x}$

$\Rightarrow y^2 = \frac{1}{4x-2+c\mathrm{e}^{-2x}}$

Substitute $y = \frac{1}{3}$ when $x = 0$:

$\left(\frac{1}{3}\right)^2 = \frac{1}{4\times0-2+c\mathrm{e}^{-2\times0}}$

$\Rightarrow \frac{1}{9} = \frac{1}{-2+c}$

$\Rightarrow c = 11$

$\therefore y^2 = \frac{1}{4x-2+11\mathrm{e}^{-2x}}$

42)

$x^3 e^y \frac{dy}{dx} = 6x^2 e^x - 3x^2 e^y$

$\Rightarrow \frac{dy}{dx} = \frac{6x^2 e^x}{x^3 e^y} - \frac{3x^2 e^y}{x^3 e^y}$

$\Rightarrow \frac{dy}{dx} = \frac{6e^x}{xe^y} - \frac{3}{x}$

Let $u = e^y \Rightarrow \frac{du}{dx} = e^y \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{1}{e^y}\frac{du}{dx} \Rightarrow \frac{dy}{dx} = \frac{1}{u}\frac{du}{dx}$

$\therefore \frac{1}{u}\frac{du}{dx} = \frac{6e^x}{xu} - \frac{3}{x}$

$\Rightarrow \frac{du}{dx} = \frac{6e^x}{x} - \frac{3u}{x}$

Rearrange into the form $\frac{du}{dx} + P(x)u = Q(x)$:

$\frac{du}{dx} + \frac{3u}{x} = \frac{6e^x}{x}$

Integrating Factor

$= e^{\int \frac{3}{x} dx}$

$= e^{3\ln|x|}$

$= e^{\ln|x^3|}$

$= |x^3|$

Let Integrating Factor $= x^3$

$\frac{du}{dx} + \frac{3u}{x} = \frac{6e^x}{x}$

$\Rightarrow x^3 \frac{du}{dx} + 3x^2 u = 6x^2 e^x$

$\Rightarrow \frac{d}{dx}(x^3 u) = 6x^2 e^x$

Use Integration by Parts:

$u = 6x^2$ $\quad$ $\frac{dv}{dx} = e^x$

$\frac{du}{dx} = 12x$ $\quad$ $v = e^x$

$\therefore \int 6x^2 e^x \, dx = 6x^2 e^x - \int 12x e^x \, dx$

Use Integration by Parts again:

$u = 12x$ $\quad$ $\frac{dv}{dx} = e^x$

$\frac{du}{dx} = 12$ $\quad$ $v = e^x$

$\therefore \int 6x^2 e^x \, dx = 6x^2 e^x - (12x e^x - \int 12 e^x \, dx)$

$= 6x^2 e^x - 12x e^x + \int 12 e^x \, dx$

$= 6x^2 e^x - 12x e^x + 12 e^x + c$

$\therefore x^3u = 6x^2\mathrm{e}^x - 12x\mathrm{e}^x + 12\mathrm{e}^x + c$

$\Rightarrow \mathrm{e}^y = \frac{6\mathrm{e}^x}{x} - \frac{12\mathrm{e}^x}{x^2} + \frac{12\mathrm{e}^x}{x^3} + \frac{c}{x^3}$

$\Rightarrow y = \ln\left(\frac{6x^2\mathrm{e}^x - 12x\mathrm{e}^x + 12\mathrm{e}^x + c}{x^3}\right)$

Substitute $y = \ln 6$ when $x = 1$:

$\ln 6 = \ln\left(\frac{6\times 1^2\times \mathrm{e}^1 - 12\times 1\times \mathrm{e}^1 + 12\times \mathrm{e}^1 + c}{1^3}\right)$

$\Rightarrow \ln 6 = \ln\left(\frac{6\mathrm{e} - 12\mathrm{e} + 12\mathrm{e} + c}{1}\right)$

$\Rightarrow \ln 6 = \ln(6\mathrm{e} + c)$

$\Rightarrow 6 = 6\mathrm{e} + c$

$\Rightarrow c = 6 - 6\mathrm{e}$

$\therefore y = \ln\left(\frac{6x^2\mathrm{e}^x - 12x\mathrm{e}^x + 12\mathrm{e}^x + 6 - 6\mathrm{e}}{x^3}\right)$

43)

$x\frac{\mathrm{d}y}{\mathrm{d}x} = y(\ln y + x)$

Let $u = \ln y \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}x} \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = y\frac{\mathrm{d}u}{\mathrm{d}x}$

$u = \ln y \Rightarrow y = \mathrm{e}^u$

$\therefore xy\frac{\mathrm{d}u}{\mathrm{d}x} = y(u + x)$

$\Rightarrow x\mathrm{e}^u\frac{\mathrm{d}u}{\mathrm{d}x} = \mathrm{e}^u(u + x)$

Rearrange into the form $\frac{\mathrm{d}u}{\mathrm{d}x} + P(x)u = Q(x)$:

$\Rightarrow x\mathrm{e}^u\frac{\mathrm{d}u}{\mathrm{d}x} = u\mathrm{e}^u + x\mathrm{e}^u$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{u\mathrm{e}^u}{x\mathrm{e}^u} + \frac{x\mathrm{e}^u}{x\mathrm{e}^u}$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{u}{x} + 1$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} - \frac{u}{x} = 1$

Integrating Factor

$= \mathrm{e}^{\int -\frac{1}{x}\mathrm{d}x}$

$= \mathrm{e}^{-\ln|x|}$

$= \mathrm{e}^{\ln|x^{-1}|}$

$= |x^{-1}|$

Let Integrating Factor $= x^{-1} = \frac{1}{x}$

$\frac{\mathrm{d}u}{\mathrm{d}x} - \frac{u}{x} = 1$

$\Rightarrow \frac{1}{x}\frac{\mathrm{d}u}{\mathrm{d}x} - \frac{u}{x^2} = \frac{1}{x}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{u}{x}\right) = \frac{1}{x}$

$\Rightarrow \frac{u}{x} = \ln x + c$

$\Rightarrow u = x\ln x + cx$

$\Rightarrow \ln y = x\ln x + cx$

$\Rightarrow y = \mathrm{e}^{x\ln x + cx}$

44)

$\frac{\mathrm{d}y}{\mathrm{d}x} = \mathrm{e}^x \cos^2 y + \sin y \cos y$

Let $u = \tan y \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \sec^2 y \frac{\mathrm{d}y}{\mathrm{d}x} \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \cos^2 y \frac{\mathrm{d}u}{\mathrm{d}x}$

$\therefore \cos^2 y \frac{\mathrm{d}u}{\mathrm{d}x} = \mathrm{e}^x \cos^2 y + \sin y \cos y$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \mathrm{e}^x + \frac{\sin y \cos y}{\cos^2 y}$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \mathrm{e}^x + \tan y$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \mathrm{e}^x + u$

Rearrange into the form $\frac{\mathrm{d}u}{\mathrm{d}x} + \mathrm{P}(x)u = \mathrm{Q}(x)$:

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} - u = \mathrm{e}^x$

Integrating Factor

$= \mathrm{e}^{\int -1 \mathrm{d}x}$

$= \mathrm{e}^{-x}$

$\frac{\mathrm{d}u}{\mathrm{d}x} - u = \mathrm{e}^x$

$\Rightarrow \mathrm{e}^{-x} \frac{\mathrm{d}u}{\mathrm{d}x} - u\mathrm{e}^{-x} = \mathrm{e}^x \mathrm{e}^{-x}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(u\mathrm{e}^{-x}) = 1$

$\Rightarrow u\mathrm{e}^{-x} = x + c$

$\Rightarrow u = (x + c)\mathrm{e}^x$

$\Rightarrow \tan y = (x + c)\mathrm{e}^x$

$\Rightarrow y = \arctan\big((x + c)\mathrm{e}^x\big)$

45) a)

$p\frac{\mathrm{d}y}{\mathrm{d}x}+qy=ry^{1-n}$

Let $u=y^n \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x}=ny^{n-1}\frac{\mathrm{d}y}{\mathrm{d}x} \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x}=\frac{1}{ny^{n-1}}\frac{\mathrm{d}u}{\mathrm{d}x} \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x}=\frac{y^{1-n}}{n}\frac{\mathrm{d}u}{\mathrm{d}x}$

$\therefore p\times\frac{y^{1-n}}{n}\frac{\mathrm{d}u}{\mathrm{d}x}+qy=ry^{1-n}$

$\Rightarrow py^{1-n}\frac{\mathrm{d}u}{\mathrm{d}x}+qny=rny^{1-n}$

$\Rightarrow p\frac{\mathrm{d}u}{\mathrm{d}x}+\frac{qny}{y^{1-n}}=\frac{rny^{1-n}}{y^{1-n}}$

$\Rightarrow p\frac{\mathrm{d}u}{\mathrm{d}x}+qny^n=rn$

$\Rightarrow p\frac{\mathrm{d}u}{\mathrm{d}x}+qnu=rn$

Rearrange into the form $\frac{\mathrm{d}u}{\mathrm{d}x}+\mathrm{P}(x)u=\mathrm{Q}(x)$:

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x}+\frac{qn}{p}u=\frac{rn}{p}$

Integrating Factor

$=\mathrm{e}^{\int\frac{qn}{p}\mathrm{d}x}$

$=\mathrm{e}^{\frac{qn}{p}x}$

$\frac{\mathrm{d}u}{\mathrm{d}x}+\frac{qn}{p}u=\frac{rn}{p}$

$\Rightarrow \mathrm{e}^{\frac{qn}{p}x}\frac{\mathrm{d}u}{\mathrm{d}x}+\frac{qn}{p}\mathrm{e}^{\frac{qn}{p}x}u=\frac{rn}{p}\mathrm{e}^{\frac{qn}{p}x}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\left(u\mathrm{e}^{\frac{qn}{p}x}\right)=\frac{rn}{p}\mathrm{e}^{\frac{qn}{p}x}$

$\Rightarrow u\mathrm{e}^{\frac{qn}{p}x}=\frac{rn}{p}\times\frac{p}{qn}\mathrm{e}^{\frac{qn}{p}x}+c$

$\Rightarrow u\mathrm{e}^{\frac{qn}{p}x}=\frac{r}{q}\mathrm{e}^{\frac{qn}{p}x}+c$

$\Rightarrow u=\frac{r}{q}+c\mathrm{e}^{-\frac{qn}{p}x}$

$\Rightarrow y^n=\frac{r}{q}+c\mathrm{e}^{-\frac{qn}{p}x}$

$\Rightarrow y=\sqrt[n]{\frac{r}{q}+c\mathrm{e}^{-\frac{qn}{p}x}}$

45) b)

As $x\to\infty$, $\mathrm{e}^{-\frac{qn}{p}x}\to 0$ so $y\to\sqrt[n]{\frac{r}{q}}$